



RZera User Manual

CSNS Data Analysis Group

RZera User Manual

The purpose of this manual is to explain to users the workflow for data reduction using RZera, and to guide users step by step through the data reduction process. If the data in your paper have been processed using RZera, please cite reference [1]. If you use and adopt the prediction results of the AI module "AIndex" in RZera in your paper, please cite reference [2]. If you use the small-angle scattering module, please cite reference [3]. If you encounter any issues during the use of the diffraction reduction function, please contact wanghao96@ihep.ac.cn. For issues during the use of the small-angle reduction function, please contact zhongjiajun@ihep.ac.cn. For collaboration inquiries, please contact durong@ihep.ac.cn.

Reference:

- [1] Wang H, Liu Z. Y, Guo H. C, Peng X. G, Xu J. P, Yin W, Zhang J. R, Du R. RZera: A customized and user-friendly software for the total scattering data reduction at CSNS[J]. *Nucl. Instrum. Methods Phys. Res., Sect. A*, 2024,1066: 169607.
- [2] Wang H, Zhong J, Li Y, Zhang J, Du R. Designing a Dataset for Convolutional Neural Networks to Predict Space Groups Consistent with Extinction Laws[J]. *Applied Crystallography*, 2025, 58(3).
- [3] Jiang H*, Zhong J*, Chen C, Zhang J, Du R. sansRZ: A python-based data reduction tool for the time-of-flight SANS instrument at CSNS[J]. *Nucl. Instrum. Methods Phys. Res., Sect. A*, 2026, 1085: 171283.

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1. Instructions for Using Powder Diffraction Spectrometer Software

1.1 Data Processing Workflow for Powder Diffraction in RZera

RZera splits the data reduction workflow for neutron powder diffraction into several sequential steps, as shown in Figure 1.1-1. In the first step, the data of the sample, sample cell, V/VNi, V/VNi cell, and empty background are each subjected to time focusing, which converts the two-dimensional histogram data into one-dimensional I-d curves. RZera provides a dedicated "Time Focusing" page for this operation. In the second step, the one-dimensional curves are used for background subtraction and vanadium normalization to obtain the reduced I-d curve of the sample. This operation is handled by the "Reduction" page in RZera. Note that some spectrometers do not require empty background data; in such cases, the data processing steps related to the empty background can be omitted. In the subsequent figures, the background refers specifically to the cell data. After that, users can continue to process the data to obtain the PDF. In RZera, there are two methods for obtaining the PDF: one is the Merge + Fourier transform method, and the other is the Hermite polynomial fitting method.

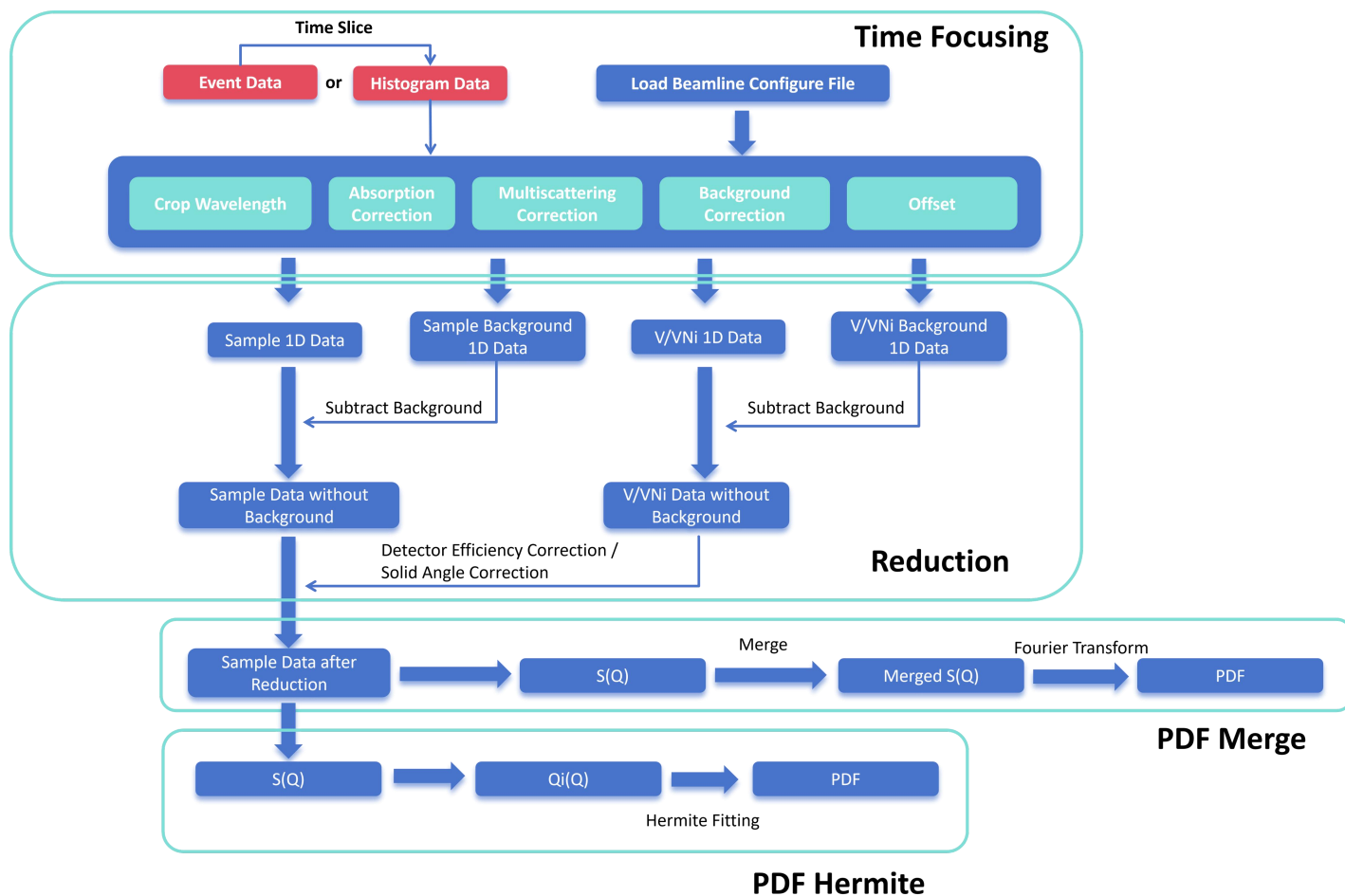


Figure 1.1-1 Data reduction workflow in RZera without empty background subtraction. In the figure, "background" refers to the cell. If empty background subtraction is required, the sample, V/VNi, and cell data must first be subtracted once by the empty background data.

In the above reduction workflow, almost every step supports visualization, allowing users to conveniently check the changes in the data throughout the entire process. However, because the steps are divided in considerable detail, the operation can be rather tedious. To address this, RZera also provides a quick reduction function: once the relevant parameters are properly configured, batch data reduction can be achieved with a single click, as shown in Figure 1.1-2. First, users still need to perform time focusing on the V/VNi data and its background data, but these two datasets can be reused across a beamtime experiment. Then, RZera provides a "Quick Reduction" page, which accepts the time-focused V/VNi and V/VNi background, as well as the pre-time-focusing sample and sample background (post-time-focusing background data are also acceptable) as inputs. After the user has set all parameters on the interface, the reduced sample data can be obtained with one click. RZera also offers a configuration saving feature: when data from different samples are collected next time, users only need to import the saved configuration, replace the sample data, and then perform the one-click

reduction again.

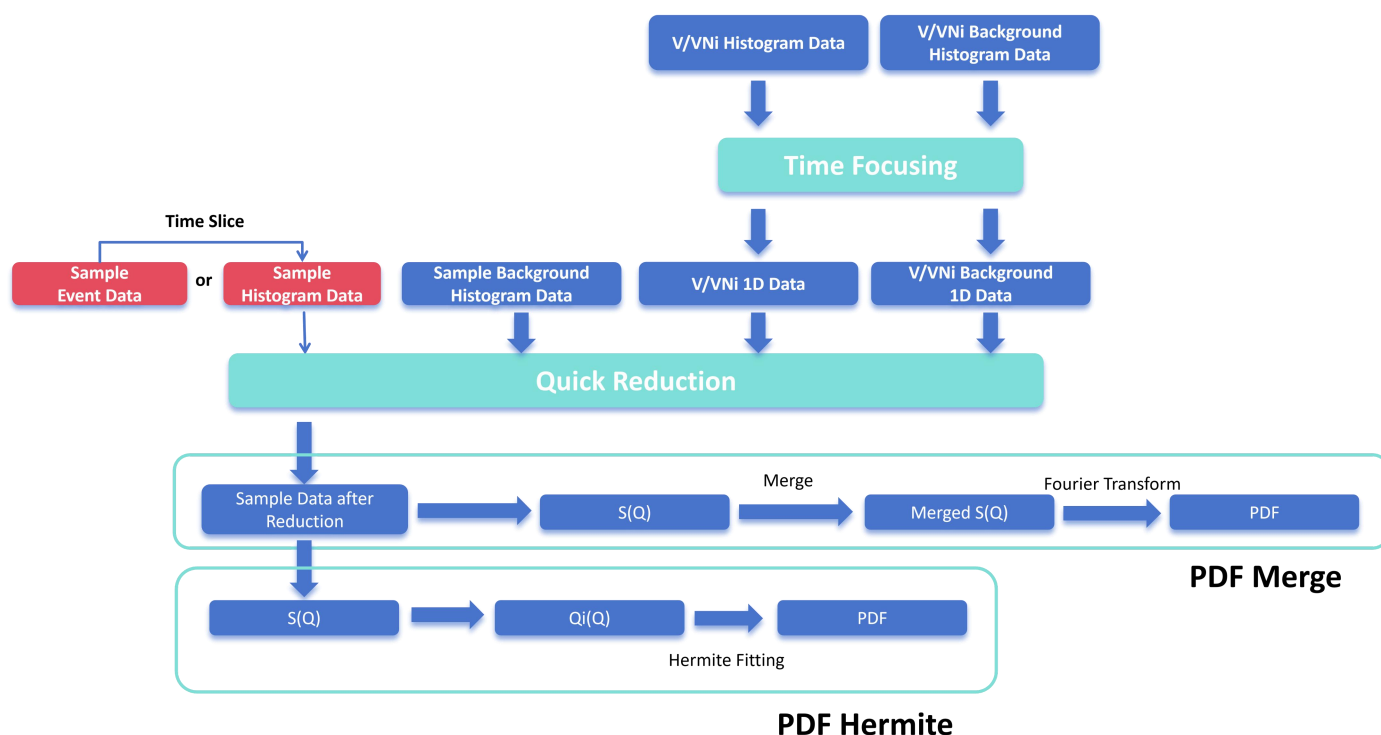


Figure 1.1-2 Quick data reduction workflow in RZera.

1.2 Data Reduction Using Histogram Data

1.2.1 File Preparation

Users need to prepare four files, namely the nxs files (provided by the spectrometer) for the V or VNi data, the V or VNi background, the sample data, and the sample background. The V or VNi background and the sample background may be the same dataset; in that case, only three files need to be prepared. As shown in Figures 1.2-1 and 1.2-2, there are three folders named after RUN numbers, which contain the nxs files for the background, V, and Si standard sample, respectively.

RUN0023566	→ the background of Si sample and V	2025/7/3 16:26	文件夹
RUN0023567	→ V	2025/5/21 13:48	文件夹
RUN0023568	→ Si Sample	2025/7/3 16:26	文件夹

Figure 1.2-1 Folders storing nxs files, with folder names corresponding to RUN numbers. The RUN number is used to label each data collection experiment.



Figure 2 Data inside a folder named by the RUN number. The nxs file is "detector.nxs" as shown in the figure, which stores the histogram data and some metadata of the experiment. There may be other data files in this folder as well, but this chapter, "Data Reduction Using Histogram Data," only uses the "detector.nxs" file.

1.2.2 Entering the RZera data reduction interface for the corresponding spectrometer

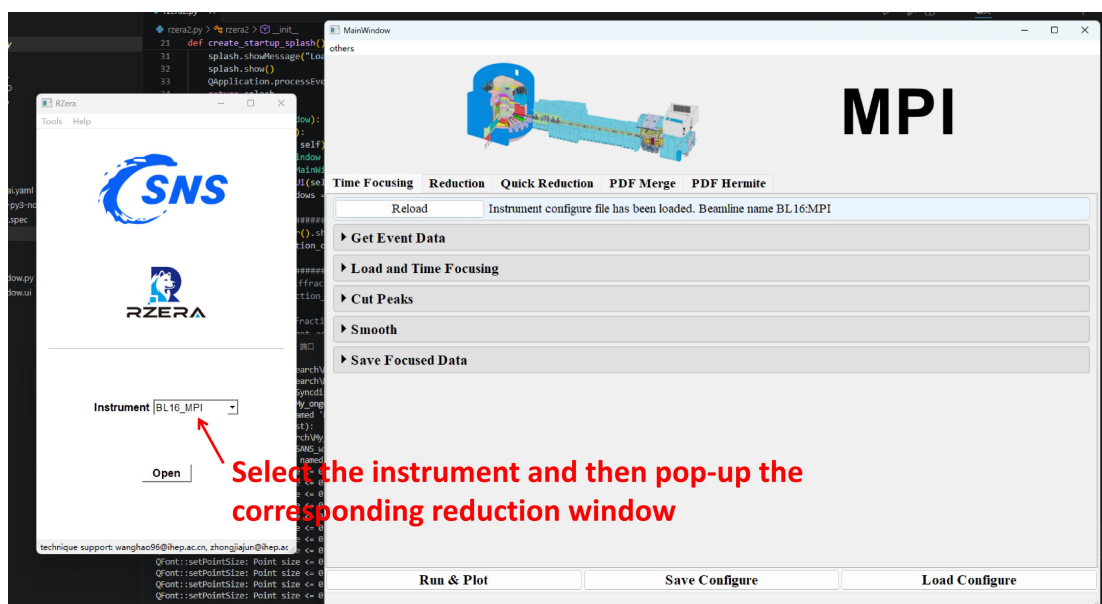


Figure 1.2-3 How to enter the data reduction window for a specified spectrometer.

As shown in Figure 1.2-3, after opening RZera, there is a drop-down box for selecting a spectrometer. Once the spectrometer used in the experiment is selected, the corresponding data reduction window for that spectrometer will pop up.

1.2.3 Performing Time Focusing on the V/VNi Data

The screenshot shows the 'Load and Time Focusing' window with the following fields and steps indicated by red arrows:

- 1. Select Histogram Data:** Points to the 'Histogram Data' tab.
- 2. Click Select Run, select target nxs file:** Points to the 'Select Run' button and the text 'RUN0023567/detector.nxs'.
- 3. Click Select Detector, select target detector:** Points to the 'Select Detector' dropdown menu showing 'group2; group3; group4; group5; group6; group7'.
- 4. Click Instru Folder, select folder with instrument files in it:** Points to the 'Instru Folder' text field containing ':search_or_work/qt_develop/rzera_offline/param_data/BL16/pidInfo'.
- 5. Check Crop Data, change wave min and wave max:** Points to the 'Crop Data' checkbox (checked) and the 'Wave Min (Å)' field (0.1) and 'Wave Max (Å)' field (4.5).
- 5. Check Carpenter Correction, make the absorption and multiple scattering correction, fill the below experimental information:** Points to the 'Carpenter Correction' checkbox (checked).
- 6. Check offset, then click "Offset Folder", select folder with offset files in it:** Points to the 'Offset' checkbox (checked) and the 'Offset Folder' text field containing ':search_or_work/qt_develop/rzera_offline/param_data/BL16/offsetfiles_RUN0023568_RT_HOT04'.

Other visible fields include 'Mask Data' (unchecked), 'Mask Folder', 'Sample Name' (V:1), 'Example: Si:1-O:2', 'Sample Mass (g)' (11.166), 'Sample Radius (cm)' (0.45), 'Beam Height (cm)' (3.0), 'Sample Height (cm)' (3.0), 'Calculated Num Density: 0.069140 Å⁻³', and 'Inputed Num Density (Å⁻³)(priority)' (0.0721). Buttons at the bottom include 'Merge Load', 'Batch Load', and 'Delete'.

Figure 1.2-4 Information input steps in "Load and Time Focusing"

In the pop-up reduction window, there is a Time Focusing page, where the first module to be operated is "Load and Time Focusing", as shown in Figure 4. When importing data following the instructions in Figure 1.2-4, several points should be noted:

(1) Regarding the "Batch Load" option: Users can import data from multiple Runs simultaneously. If "Batch Load" is checked, RZera will perform time focusing on each of the imported datasets separately. However, if it is not checked, RZera will merge the data from these Runs and then perform time focusing on the merged data.

(2) Regarding "Instru Folder", RZera will provide this.

(3) Regarding "Sample_Name" under "Carpenter Correction": Enter the ratio according to the actual molar fraction of the V:Ni sample. If the exact number density value is known, instead of using the calculated value, you may check "Inputed Num Density" and manually enter the accurate value.

(4) Regarding "Offset Folder", RZera or the spectrometer team can provide this.

After completing the information input, click "Load", and the software will begin loading the data and performing time focusing based on the entered information. The loaded data will then be saved in the list at the bottom of this module, as shown in Figure 1.2-5.

Merge Load Batch Load Delete			
	Name	Detector	Select
1	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:16)	group2	☐
2	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:20)	group3	☐
3	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:24)	group4	☐
4	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:31)	group5	☐

Figure 1.2-5 Data obtained after loading.

After the data are loaded, this page also provides peak removal and smoothing functional modules, located below the "Load and Time Focusing" module, as shown in Figure 1.2-6. These modules are collapsed by default and are used to remove unwanted diffraction peaks from the data. For vanadium, we need to remove all of its diffraction peaks.

	Name	Detector	Select
3	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:24)	group4	☐
4	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:31)	group5	☐
5	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:35)	group6	☐
6	RUN0023567_crop_carpenterCorrection_offset_time(2025-05-28-15:59:39)	group7	☐

► Cut Peaks

► Smooth

► Save Data

Figure 1.2-6 Peak removal and smoothing functional modules.

After clicking "Cut Peaks", the module will expand, as shown in Figure 1.2-7. This module consists of two parts: the upper part is used to automatically calculate the positions of diffraction peaks to be removed based on a CIF file, and the lower part is for manually locating diffraction peak positions.

Cut peaks with cif

▼ Cut Peaks
☐ Plot

Add Cif
Delete
Check
Add Peaks

Name	a	b	c	α	β	γ	A	B

Add Default Peak
Add Peak
Delete

Detector	Peak Left	Peak Right	Select
Cut peaks manually			

Figure 1.2-7 The "Cut Peaks" functional module.

RZera has built-in default peak positions for vanadium. Simply click "Add Default Peak" in the manual peak removal section to automatically import the V peak positions, as shown in Figure 1.2-8.

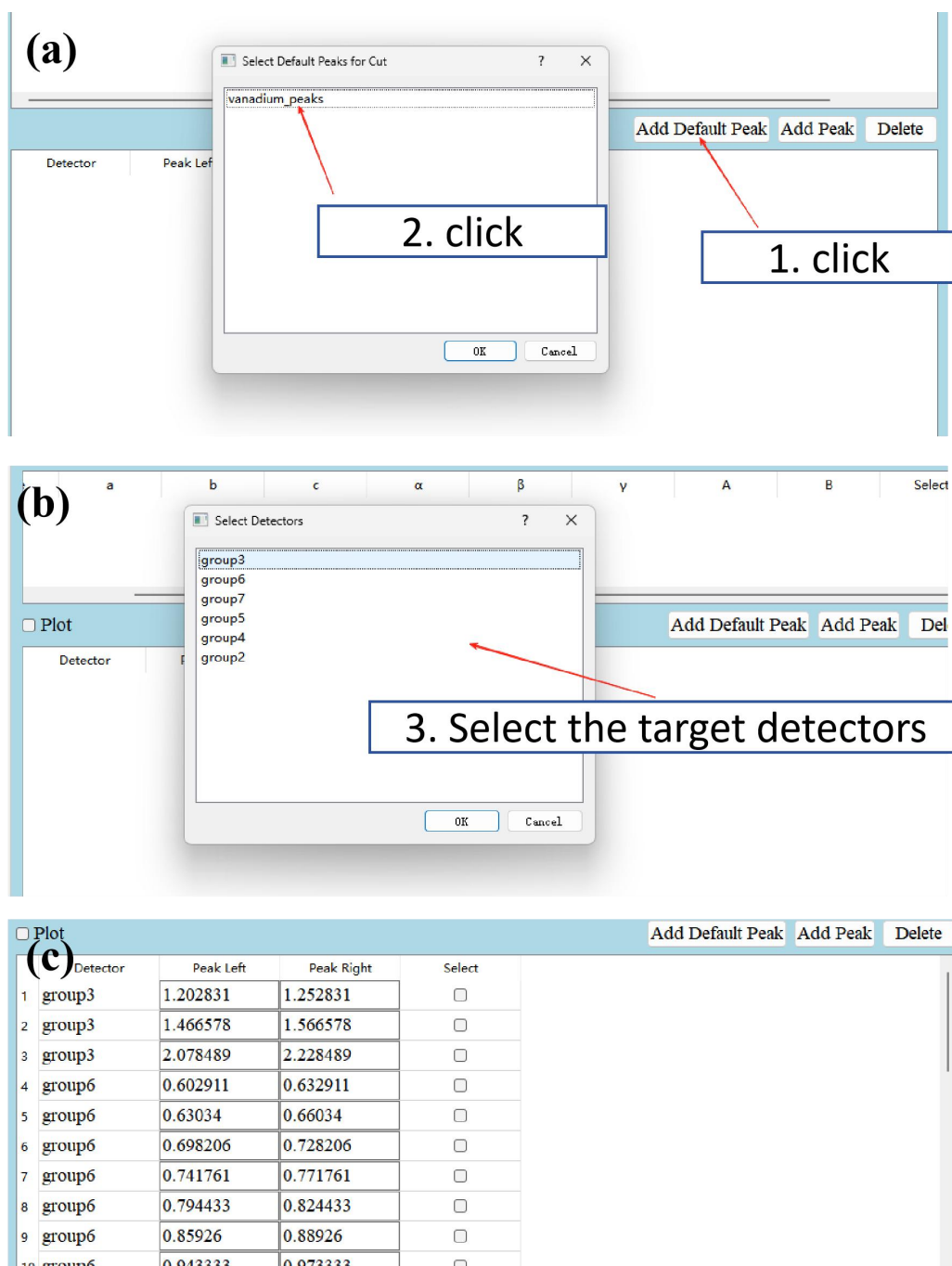


Figure 1.2-8 Importing default V peaks. (a) After clicking the button, select the type of default peaks to import. (b) Select the detector for which the peak information is to be imported. The detectors in this list are consistent with those selected during the previous data loading step. (c) The peak position information displayed after import.

After importing, check the "plot" boxes in both the "Load and Time Focusing" module and the "Cut Peaks" module, then click the "Run & Plot" button at the bottom of the entire page. The "Run & Plot" button will execute the enabled functional modules in order from top to bottom. Additionally, if the "plot" box in a module is checked, the data processed by that module will be added to the plotting window and the plotting

window will pop up, enabling data visualization, as shown in Figure 1.2-9.

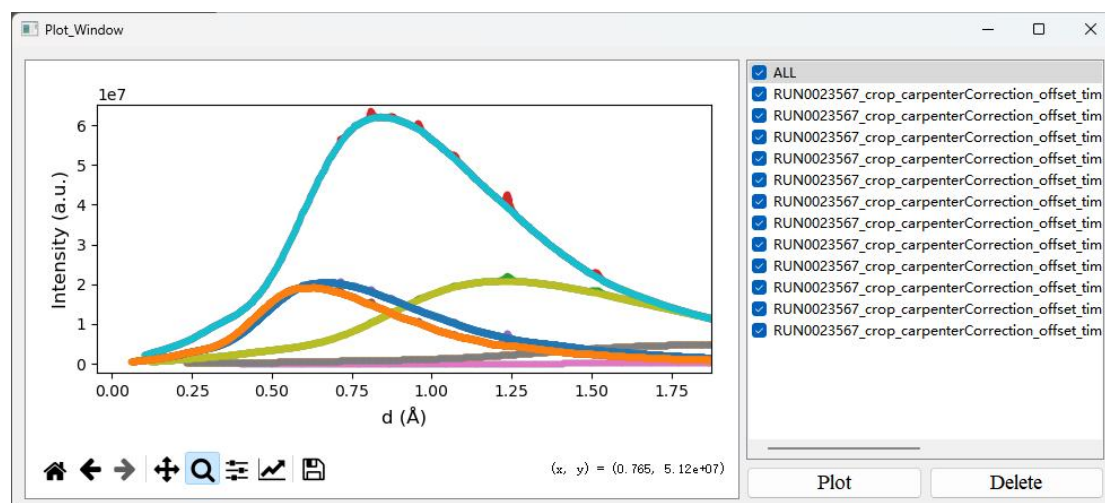


Figure 1.2-9 Comparison of data before and after peak removal. The reason for this comparison is that the software executes modules from top to bottom: first, "Load and Time Focusing" is executed, and since the "plot" box in this module is checked, the data before peak removal are saved in the plotting window. Then, "Cut Peaks" is executed, and after peak removal, since the "plot" box in this module is also checked, the data after peak removal are saved in the plotting window.

After peak removal, the removed portions are filled with straight lines. RZera provides the "Smooth" module (Figure 1.2-6) to smooth the data, and its expanded view is shown in Figure 1.2-10. Clicking "Add Default Smooth" within this module allows you to apply the built-in smoothing parameters of RZera with one click.

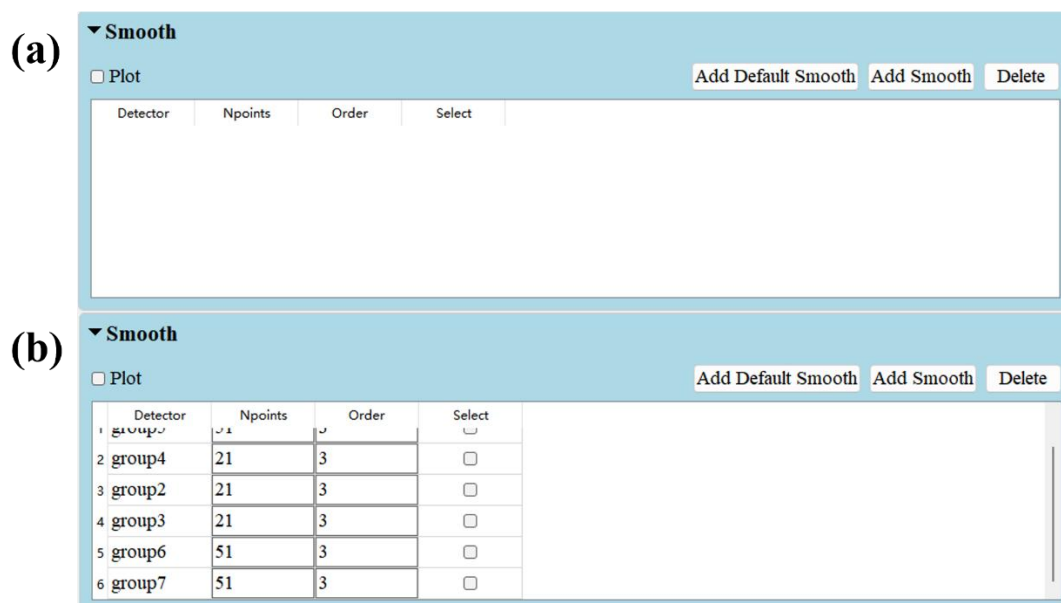


Figure 1.2-10 (a) The expanded interface of the "Smooth" module. (b) The interface after clicking "Add Default Smooth".

After checking the "plot" box in the "Smooth" module as well, click the "Run & Plot" button to achieve a comparison of the data before and after smoothing, as shown in Figure 1.2-11.

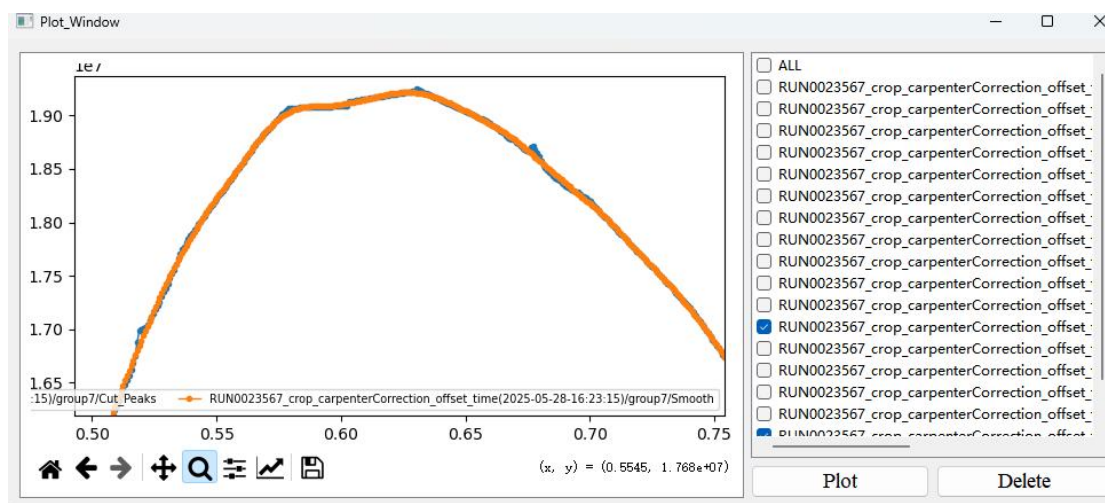


Figure 1.2-11 Comparison of data from detector group 7 before and after smoothing. It can be seen that the data are significantly smoother after smoothing.

After smoothing, expand the final "Save Data" module (as shown in Figure 1.2-12) to prepare for saving the processed data. After configuring the "Save Data" module, click "Run & Plot" again, and the processed data will be saved to the specified path.

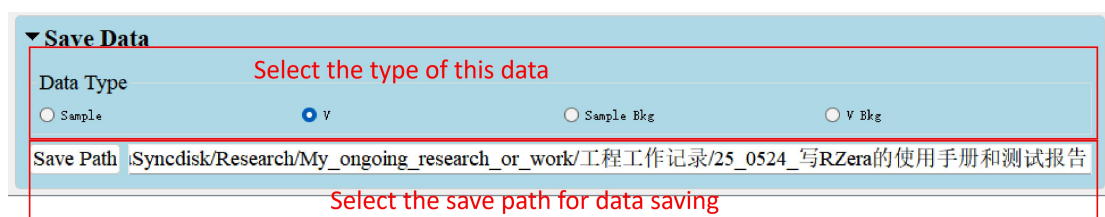


Figure 1.2-12 The interface of the "Save Data" module. The "Data Type" must be consistent with the type of data currently being processed. In this example, we are processing V or VNi sample data, so "V" should be selected. Then click "Save Path" to specify the save location.

The processed file is a ".nc" file, which stores the processed data and the processing history internally, as shown in Figure 1.2-13.

	v_RUN0023567_crop_carpeneterCorrection_offset_time(2025-05-28-173126)_group2.nc	2025/5/28 17:33
	v_RUN0023567_crop_carpeneterCorrection_offset_time(2025-05-28-173130)_group3.nc	2025/5/28 17:33
	v_RUN0023567_crop_carpeneterCorrection_offset_time(2025-05-28-173135)_group4.nc	2025/5/28 17:33
	v_RUN0023567_crop_carpeneterCorrection_offset_time(2025-05-28-173146)_group5.nc	2025/5/28 17:33
	v_RUN0023567_crop_carpeneterCorrection_offset_time(2025-05-28-173151)_group6.nc	2025/5/28 17:33
	v_RUN0023567_crop_carpeneterCorrection_offset_time(2025-05-28-173157)_group7.nc	2025/5/28 17:33

Figure 1.2-13 Saved Data

1.2.4 Performing Time Focusing on the V/VNi Background

The operation procedure for this step is the same as in the previous step (Step 3), but there are several differences or points to note, as follows:

- (1) The nxs file to be loaded should be replaced with the background

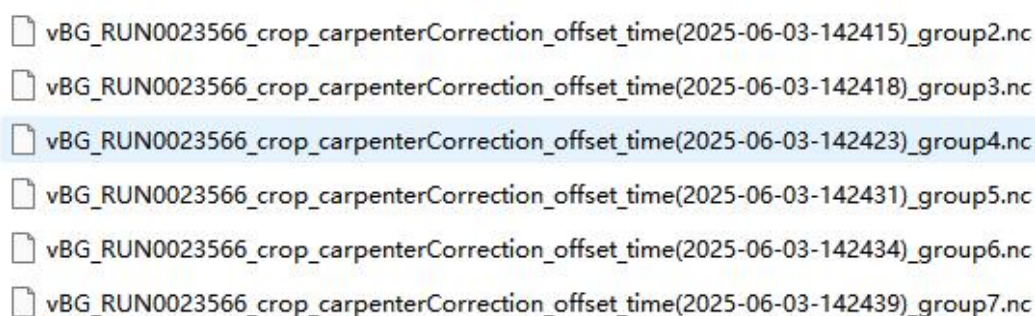
corresponding to V/VNi.

(2) If Carpenter Correction was applied to the V/VNi data during "Load and Time Focusing", it is recommended to apply exactly the same Carpenter Correction to the background data as well, in order to cancel out its effect during the subsequent background subtraction.

(3) Since the background may not contain diffraction peaks, the Cut Peaks and Smooth steps may not be necessary, depending on the actual situation.

(4) When saving the data at the end, "Data Type" should be set to "V Bkg".

The final saved data are shown in Figure 1.2-14.



vBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-142415)_group2.nc
vBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-142418)_group3.nc
vBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-142423)_group4.nc
vBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-142431)_group5.nc
vBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-142434)_group6.nc
vBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-142439)_group7.nc

Figure 1.2-14 The saved V/VNi background data.

1.2.5 Performing Time Focusing on the Sample

The operation procedure for this step is the same as Step 3, but there are several differences or points to note, as follows:

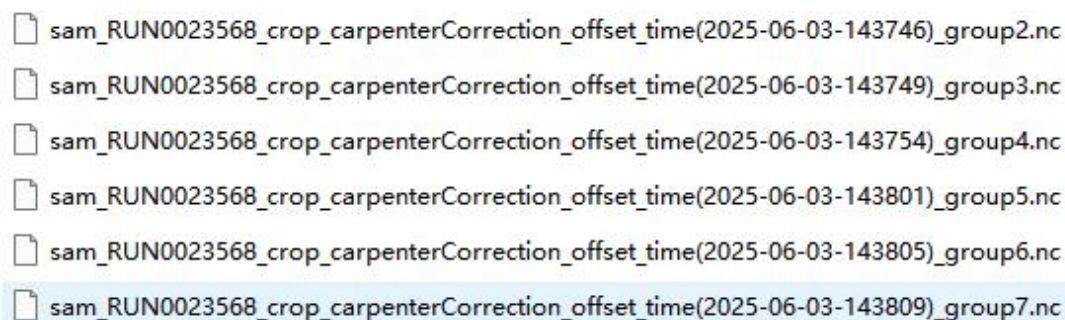
(1) The nxs file to be loaded should be replaced with that of the sample.

(2) Carpenter Correction is not necessarily required for the sample. If only diffraction data are needed, it can be skipped. However, if further processing to obtain the PDF is required, then it is needed. If the correction is to be applied, it is necessary to ensure that the sample is either a cylinder or a powder contained in a cylindrical container, and that all relevant input parameters are entered correctly; otherwise, the correction may be inaccurate.

(3) Since the sample may not contain extraneous diffraction peaks, the Cut Peaks and Smooth steps may not be needed, depending on the specific situation.

(4) When saving the data at the end, "Data Type" should be set to "Sample".

The final saved data are shown in Figure 1.2-15.



```

sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143746)_group2.nc
sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143749)_group3.nc
sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143754)_group4.nc
sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143801)_group5.nc
sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143805)_group6.nc
sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143809)_group7.nc

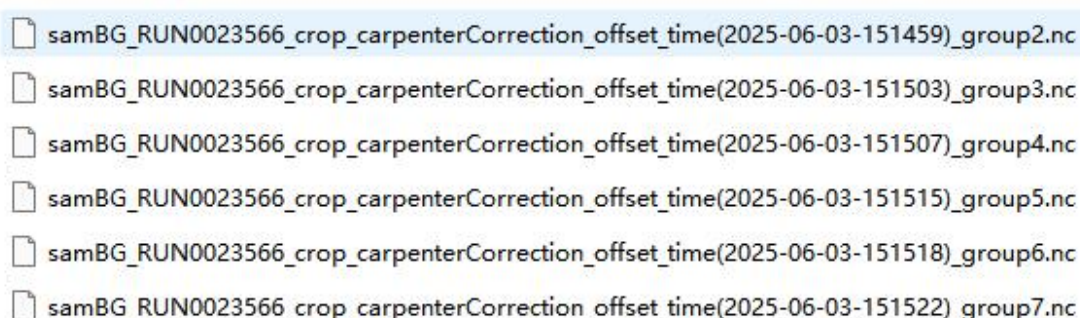
```

Figure 1.2-15 The saved sample data.

1.2.6 Performing Time Focusing on the Sample Background

The operation procedure for this step is the same as Step 3, but there are several differences or points to note, as follows:

- (1) The nxs file to be loaded should be replaced with that of the sample background.
 - (2) If Carpenter Correction was applied to the sample during "Load and Time Focusing", it is recommended to apply exactly the same Carpenter Correction to the background data as well, in order to cancel out its effect during the subsequent background subtraction.
 - (3) Since the background may not contain diffraction peaks, the Cut Peaks and Smooth steps may not be necessary, depending on the actual situation.
 - (4) When saving the data at the end, "Data Type" should be set to "Sample Bkg".
- The final saved data are shown in Figure 1.2-16.



```

samBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-151459)_group2.nc
samBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-151503)_group3.nc
samBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-151507)_group4.nc
samBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-151515)_group5.nc
samBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-151518)_group6.nc
samBG_RUN0023566_crop_carpenterCorrection_offset_time(2025-06-03-151522)_group7.nc

```

Figure 1.2-16 The saved sample background data.

1.2.7 Using the data obtained in Sections 1.2.3–1.2.6 to obtain the reduced diffraction pattern.

After completing Steps 3–6, we have obtained the time-focused data for four types: sample, sample background, V/VNi, and V/VNi background. Next, we need to switch

the tab page from "Time Focusing" to "Reduction". The interface of the Reduction page is shown in Figure 1.2-17.

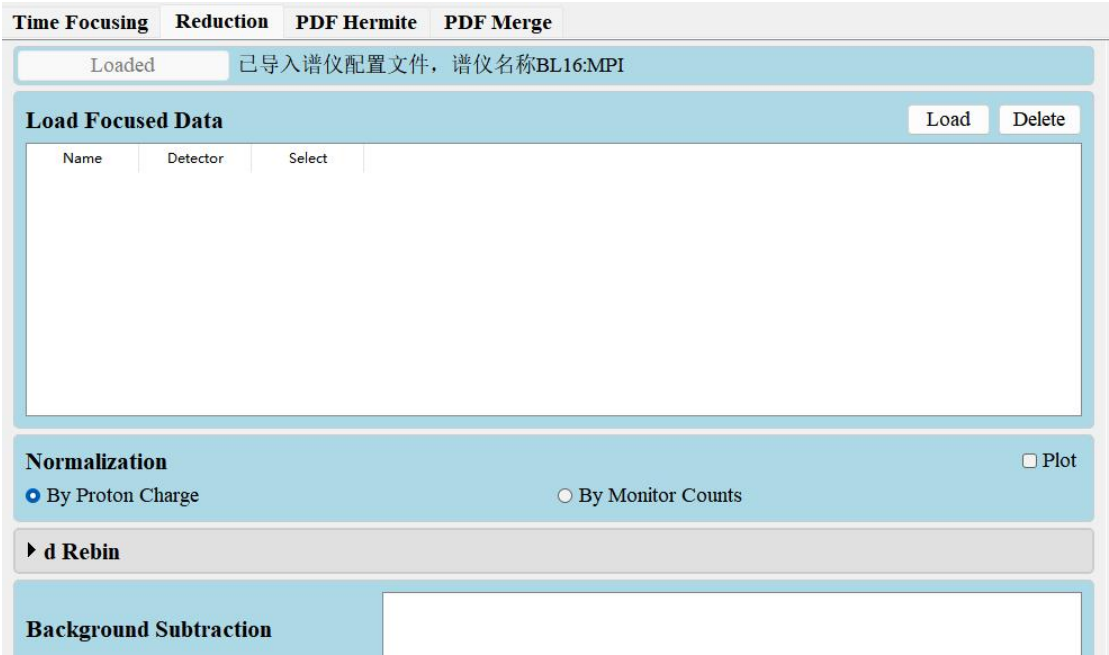


Figure 1.2-17 The "Reduction" tab page.

First, we operate the "Load Focused Data" module. Click "Load" to import all the data obtained in Steps 3–6, as shown in Figure 1.2-18.

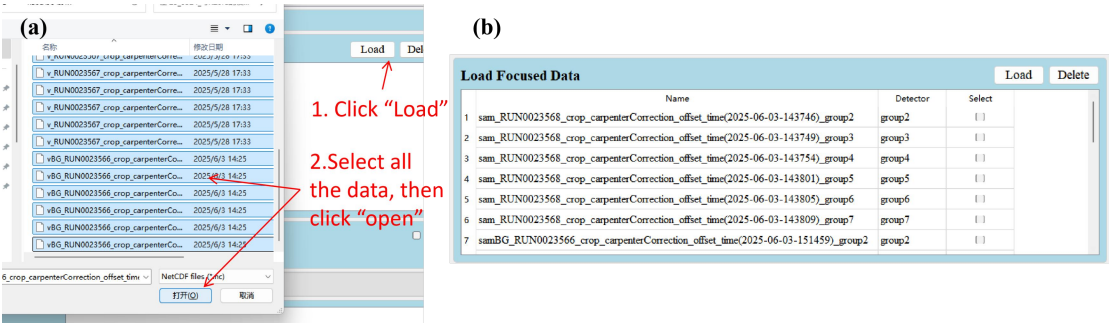


Figure 1.2-18 (a) Schematic diagram of the "Load Focused Data" module operation. (b) The "Load Focused Data" module after data loading.

Next, consider the "Normalization" module, which offers two options: "By Proton Charge" and "By Monitor Counts". These refer to normalizing the data by proton charge and by monitor counts, respectively. In theory, either choice is correct; however, there may be cases where a monitor was not used during the experiment, in which case selecting "By Monitor Counts" will cause the program to report an error. Therefore, the appropriate option should be chosen based on the actual experimental conditions.

Next, expand the "d_Rebin" module and click "Add Default Rebin". After selecting the detectors to be processed in the pop-up window, the module will load the

default d_rebin parameters for each detector, as shown in Figure 1.2-19. After loading the default parameters, users can also modify the d_rebin parameters directly in the interface.

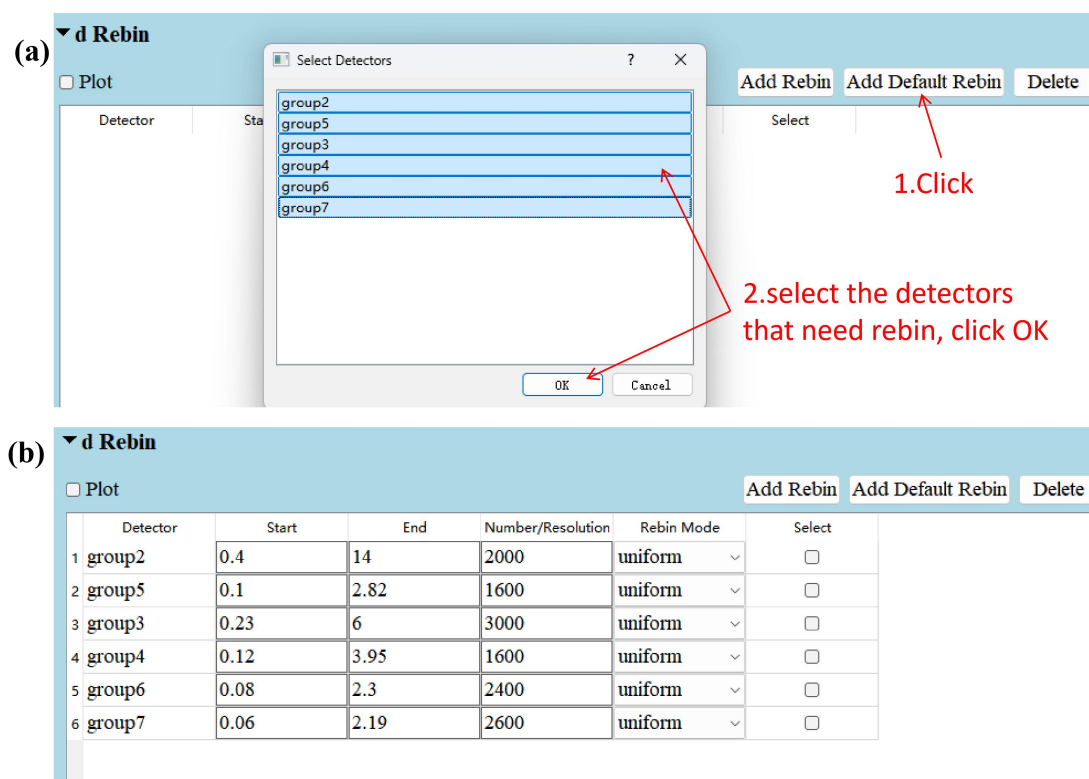


Figure 1.2-19 (a) The workflow for operating the "d_Rebin" module. (b) The "d_Rebin" module after loading the default parameters.

Next, operate the subsequent "Background Subtraction" module as shown in Figure 1.2-20. This module is dedicated to background subtraction. It can be noted that there are multiple consecutive "Background Subtraction" modules in the interface, which are used for subtracting the background from the sample and V/VNi data respectively (there is no required order for the background subtraction). Click "Load Sample" to load the sample or V/VNi data, and then click "Load Background" to load the corresponding background. After loading, the module provides real-time display of the data after background subtraction. The "Background Scale" in the module refers to the scale factor multiplied to the background before subtraction. Sliding the "Background Scale" slider allows adjustment of this factor, and the changes in the data after applying the factor are reflected in real time in the plot on the right. The default scale factor is 1, but there may be cases where the background is higher than the data; in such cases, the scale factor must be adjusted to ensure that all data remain positive after background subtraction. For guidance on which value to choose as the scale factor, please consult the spectrometer scientist for the corresponding instrument.

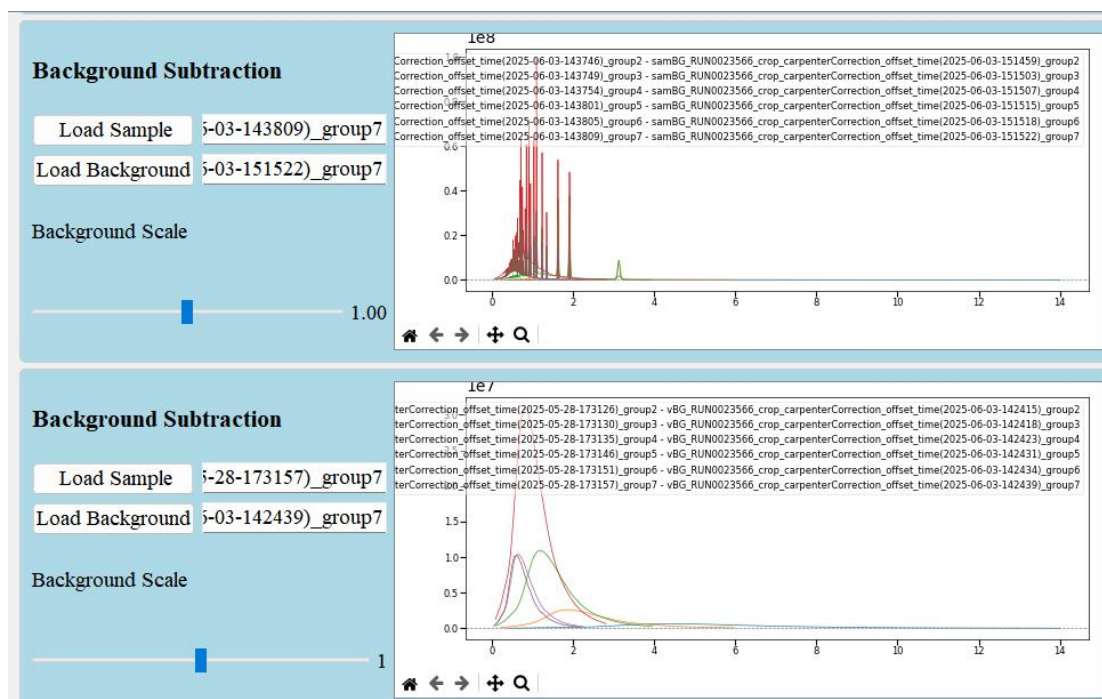


Figure 1.2-20 The interface after loading the sample, V/VNi, and their corresponding backgrounds. After the data are loaded, the right panel will display the background-subtracted data. If the data do not appear, you can click the "home" button (the small black house icon) in the lower-left corner of the plotting area, and the data will be displayed normally.

Next, operate the subsequent "Division" module. Click "Load Sample" to import the sample data, and click "Load Calibration" to import the V/VNi data. The data imported here will be divided after having undergone background subtraction from the preceding modules. The module after data import is shown in Figure 1.2-21.

Division		☐ Plot
Load Sample	5-143805_group6; sam_RUN0023568_crop_carpen...Correction_offset_time(2025-06-03-143809)_group7	
Load Calibration	28-173151_group6; v_RUN0023567_crop_carpen...Correction_offset_time(2025-05-28-173157)_group7	

Figure 1.2-21 The "Division" module after importing the sample and V/VNi data.

Next is a skippable module, "AIndex", which is developed based on research results published in 2025 by the China Spallation Neutron Source data analysis group. Its function is AI-based prediction of space groups and unit cell parameters. Users only need to import the reduced data to predict the space group and unit cell parameters with one click, and can easily check whether the prediction results are reliable.

First, click the "Load Pattern" button to import the reduced sample data, as shown in Figure 1.2-22 (a)(b). Then, click the "Prediction" button, and the software will call the trained deep neural network to provide prediction results for the crystal system,

space group, and unit cell parameters corresponding to each diffraction pattern, which are displayed in the list panel on the right, as shown in Figure 1.2-23. The panel contains two columns: the first column shows the prediction results, and the second column shows the confidence level of each prediction. By selecting a prediction result in the list panel and clicking the "Check" button, AIndex will calculate the diffraction peak positions using the predicted results and display them together with the diffraction pattern, as shown in Figure 1.2-23, allowing users to conveniently verify the accuracy of the prediction.

There are certain limitations to using this module, as well as specific requirements for the data: (1) Currently, AIndex can only perform predictions for cubic, tetragonal, trigonal, and hexagonal crystal systems. (2) The data must be from a single-phase sample; otherwise, correct prediction cannot be performed. (3) The unit cell parameters of the sample should be within the range of 3–15 Å; otherwise, correct prediction cannot be performed. (4) The data should ideally cover the range 0.5–10 Å, or it must be ensured that no diffraction peaks exist in the uncovered regions; otherwise, the prediction results are likely to be inaccurate.

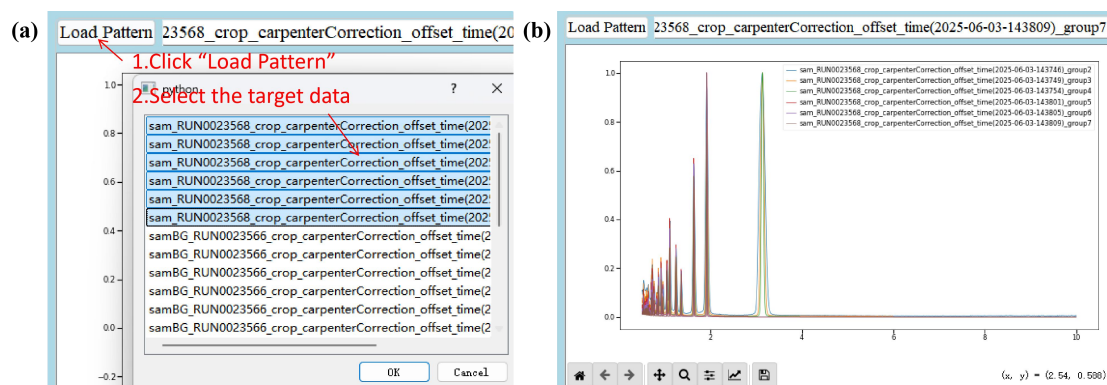
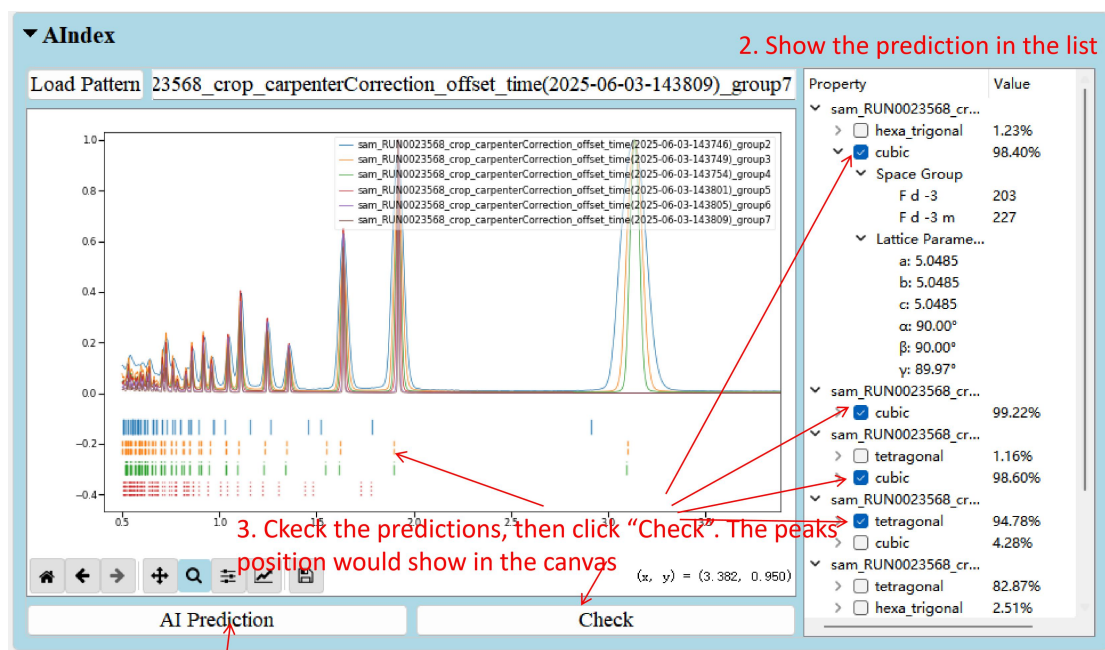


Figure 1.2-22 (a) The workflow for importing data into "AIndex". (b) After data import, AIndex reorganizes the data into a diffraction pattern consisting of 4196 equally spaced points over the range 0.5–10 Å and displays it. This reorganization is intended to match the input format required by the neural network and is used only internally within this module; it does not affect the final reduction results.



1. Click "AI Prediction".

Figure 1.2-23 The workflow for predicting space group and unit cell parameters using AIndex with the assistance of AI.

Then, operate the "Save Reduced Data" module. This module is very simple: after expanding it, just click "Save Path" to specify the save location for the data. Finally, click "Run & Plot", and the software will process the modules one by one from top to bottom. The processed data will then be saved to the path specified in the "Save Reduced Data" module, as shown in Figure 1.2-24.

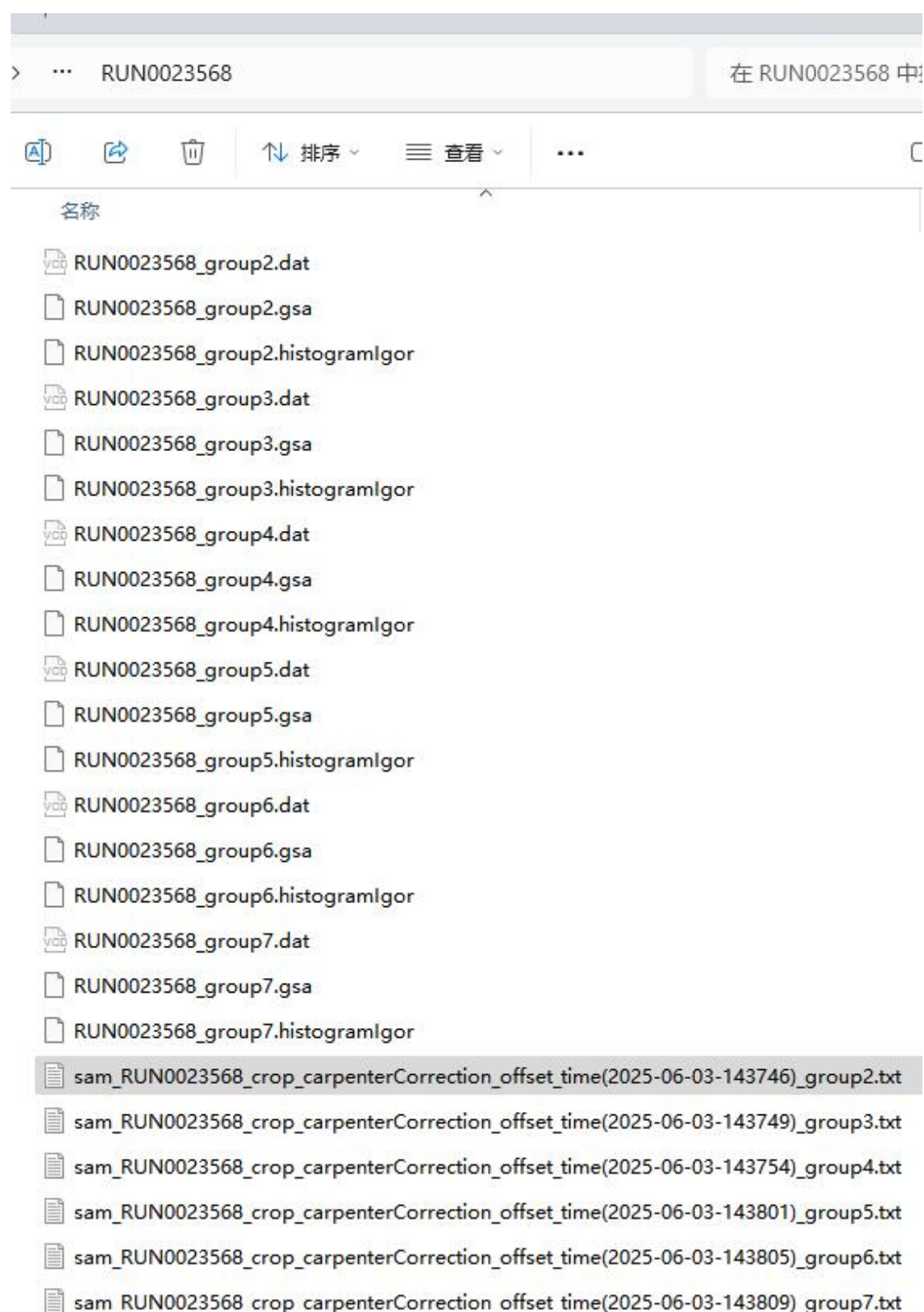


Figure 1.2-24 The saved reduced data. For each dataset, four files are saved: time-of-flight data files for data processing in the three software packages "FullProf", "GSAS", and "Z-Retiveld", as well as a txt file containing the d-spacing data.

1.2.8 Quick Reduction

After obtaining the time-focused data of V/VNi and its cell (container) through Sections 1.2.3 and 1.2.4, quick data reduction can be performed on the Quick Reduction page in RZera. The following figure shows the interface of Quick Reduction along with configuration and usage instructions.

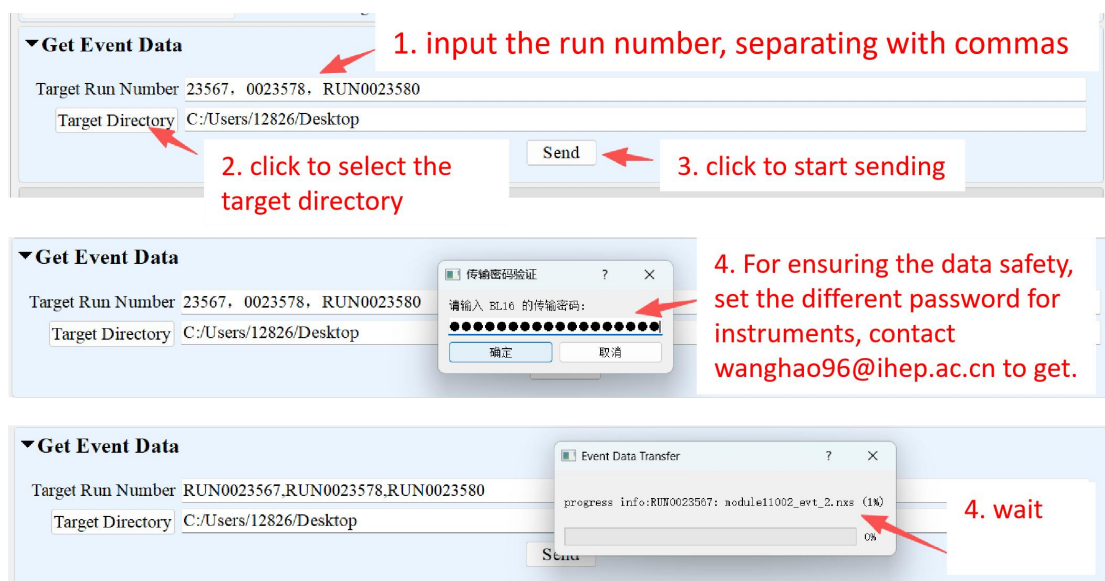
The screenshot shows the 'MPI' Quick Reduction interface. At the top, there's a 'MainWindow' title bar and a 'Reload' button. Below it, a status bar says 'Instrument configure file has been loaded. Beamline name BL16:MPI'. The main section is titled 'Quick Reduction' and contains several tabs: 'Histogram Sample', 'Event Sample', 'Select Run', 'Batch Load', 'Select Sample Cell', 'Select Van', 'Select Van Cell', 'Select Detector', 'Instru Folder', 'Mask Data', 'Crop Data', 'Sample Correction', 'Offset', 'Normalization Methods', 'Bkg Subtraction', and 'Plot'. Red arrows point to various fields with instructions: 'select the sample runs (nxs)' points to 'Select Run'; 'if check, deal the runs seperately, otherwise sum them before dealing' points to 'Batch Load'; 'sample cell data, nxs/nc optional' points to 'Select Sample Cell'; 'V/VNi nc data after time focusing' points to 'Select Van'; 'V/VNi cell nc data after time focusing' points to 'Select Van Cell'; 'target detectors' points to 'Select Detector'; 'folder with instrument files in it' points to 'Instru Folder'; 'check it for wave truncation' points to 'Wave Min (Å) 0.1'; 'folder with offset files in it' points to 'Offset Folder'; 'Normalization Methods' points to 'By Proton Charge' and 'By Monitor Counts'; 'Scale factor of sample and V/VNi cell' points to 'Sample Bkg Scale 1.0' and 'Van Bkg Scale 0.5'; and 'click the "reduction" finally' points to the 'Reduction' button. The bottom of the interface has three buttons: 'Run & Plot', 'Save Configure', and 'Load Configure'.

1.3 利用事例数据进行数据规约

事例数据和直方图数据最大的不同是其保存了每个中子计数的时间戳。因此，可以对事例数据进行时间切割，以观察同一个 RUN 下不同时间段采集的衍射谱。该功能对原位实验来说比较方便。

1.3.1 获取事例数据到本地

在 Time Focusing 页面，第二个模块是“Get Event Data”模块，使用该模块，可以将指定 RUN（可多个）的示例数据传输到 RZera 所在的服务器，用于进行数据归约。传输方式如下图：



Get Event Data 模块使用示意图

1.3.2 利用事例数据进行规约

和第一节的利用直方图数据相比，其流程是一样的，但是有下面两点不同，需要注意：

（1）需要准备的文件不同。

在需要准备的样品、V/VNi、样品背底、V/VNi 背底这 4 类文件中，只有样品需要准备的不是直方图数据了，而是事例数据。其他类则和之前完全一样。样品的直方图数据文件如图 25 所示。RUN 号文件夹下必须有的文件是每个模块的示例数据文件“module*_evt_*.nxs”，以及“pcLog.nxs”文件。并且，一定要保证每个模块的示例数据文件的数量都是相同的。在图 25 中，每个模块都有 1 个示例数据文件，对于其他的 RUN，可能会存在每个模块有多个示例数据文件的情况，无论如何请保证不同模块文件数量相同。

... BL15 > Hutch1 > RUN0001836 > 在 RUN0001836 中搜 Q

排序 查看 ... 预览

名称	修改日期	类型	大小
module11507_evt_0.nxs	2025/5/16 15:32	NXS 文件	16,407 KB
module11508_evt_0.nxs	2025/5/16 15:32	NXS 文件	16,440 KB
module11509_evt_0.nxs	2025/5/16 15:32	NXS 文件	16,679 KB
module11510_evt_0.nxs	2025/5/16 15:32	NXS 文件	17,064 KB
module11511_evt_0.nxs	2025/5/16 15:32	NXS 文件	18,001 KB
module11601_evt_0.nxs	2025/5/16 15:32	NXS 文件	18,162 KB
module11602_evt_0.nxs	2025/5/16 15:32	NXS 文件	19,481 KB
module11603_evt_0.nxs	2025/5/16 15:32	NXS 文件	23,606 KB
module11604_evt_0.nxs	2025/5/16 15:32	NXS 文件	27,143 KB
module11605_evt_0.nxs	2025/5/16 15:32	NXS 文件	28,561 KB
module11606_evt_0.nxs	2025/5/16 15:32	NXS 文件	29,556 KB
module11607_evt_0.nxs	2025/5/16 15:32	NXS 文件	31,539 KB
module11608_evt_0.nxs	2025/5/16 15:32	NXS 文件	32,285 KB
module11609_evt_0.nxs	2025/5/16 15:32	NXS 文件	33,511 KB
module11610_evt_0.nxs	2025/5/16 15:32	NXS 文件	34,708 KB
module11611_evt_0.nxs	2025/5/16 15:32	NXS 文件	36,176 KB
module11612_evt_0.nxs	2025/5/16 15:33	NXS 文件	35,807 KB
module11613_evt_0.nxs	2025/5/16 15:33	NXS 文件	33,655 KB
pcLog.nxs	2025/5/16 15:49	NXS 文件	1,776 KB

图 25 高压谱仪 RUN0001836 号实验采集的事例数据

(2) “Time Focusing”页面中导入数据的方式不同

“Time Focusing”页面导入示例数据的界面如图 26 所示。数据仍然在“Load and Time Focusing”模块中导入，首先需要在该模块中切换到“Event Data”这个 tab，然后点击“Select Event Data”这个按钮，选择保存事例数据的文件夹即可。请注意事例数据目前只支持导入一个 RUN 的数据，不支持多个 run 批量导入和切割。数据导入后，软件会自动读取该 RUN 的总测试时间并显示在界面上 (Exp Time)，如图 26 (b) 所示。用户可设置时间切割后得到的直方图数据中的时间间隔（默认 8us），以及最重要的，可以设置时间切割参数。如图 26 (b) 所示，时间切割参数是一个 3 列的列表，可以通过“Add”“Remove”来增删行。每行是一组 3 个参数，分别指切割起始时间，切割结束时间，以及这段时间要等间隔切的段数。在图 26 (b) 中输入的切割参数，意味着要在 0-10min 切一段，10-20min 切 2 段，这意味着最后会切出 0-10，10-15，15-20 三段数据。

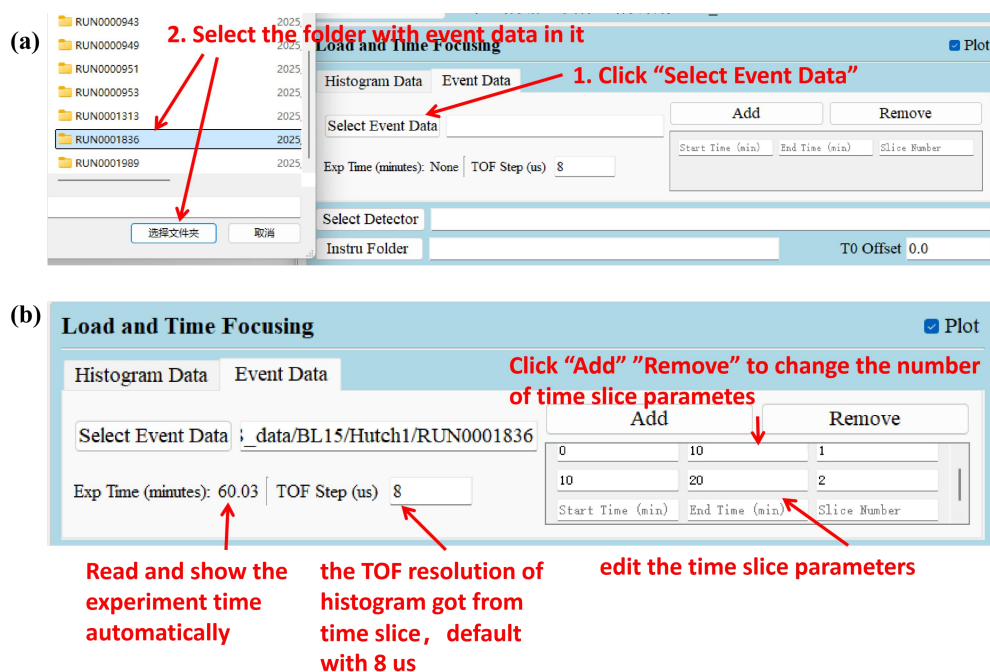


图 26 “Load and Time Focusing”导入事例数据的操作说明

1.4 利用 Time Focusing 后的数据获得 PDF (Merge/Hermite)

用户如果需要获取 PDF 数据，首先需要按照第一节步骤获取四类文件的 Time Focusing 后的数据文件，然后根据本节的指引进行处理。RZera 提供两种获取 PDF 数据的方法，分别是对衍射谱进行 Merge 后获取 PDF 数据，以及通过 Hermite 多项式拟合的方式获取 PDF 数据。下面分别介绍如何操作：

1.4.1 Merge 后获取 PDF 数据

Step1: 切换到“PDF Merge”页面并完成数据规约

用户首先需要切换到“PDF Merge”页面，然后会发现该页面最开始的几个模块和“Reduction”页面中，完全一样。这是因为在获取 PDF 数据之前要先对数据进行 reduction。这里，知道“Division”模块，其操作方式和第一节介绍的是完全一样的，这里不做赘述。

Step2: 将规约后的 I(d)数据转换为 S(Q)数据。

接下来，用户需要操作后续的“Convert to S(Q)”模块，将 I(d)数据转换成 S(Q)数据，如图 27 所示。

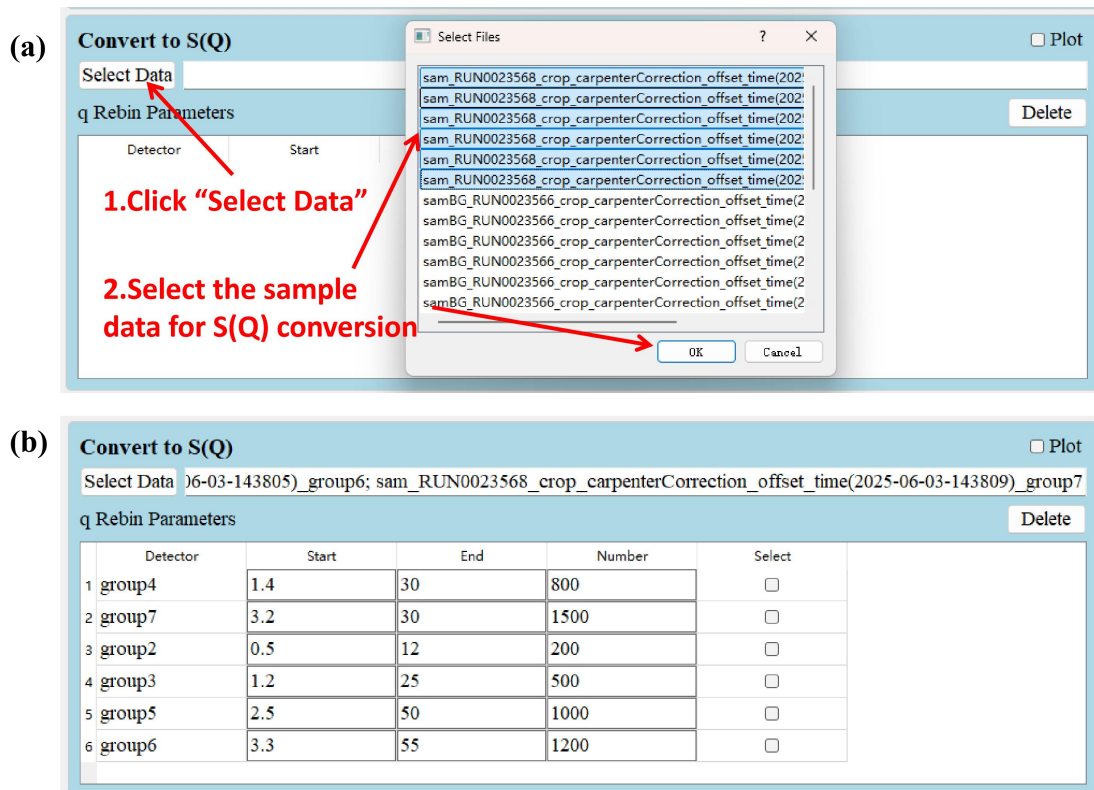


图 27 (a)选择需要转 S(Q)的样品数据的操作流程。(b)选择完成后，界面上会自动出现选择的数据所关联探测器的 q Rebin 默认参数，这里是可修改的，用户可以根据数据真实情况进行调整。

根据图 27 中的指示处理好数据后，强烈建议用户先对 S(Q)数据进行评估后再继续后续流程。数据评估的方式为先勾选“Convert to S(Q)”中的“Plot”，然后点击页面最下方的“Run & Plot”，软件会逐个执行模块并对勾选绘图的软件进行绘图，绘图的结果如图 28 所示。图中显示来自 group2 和 group4 的数据有问题，经过调节 q rebin 的参数后，问题得到改善，如图 29 所示。

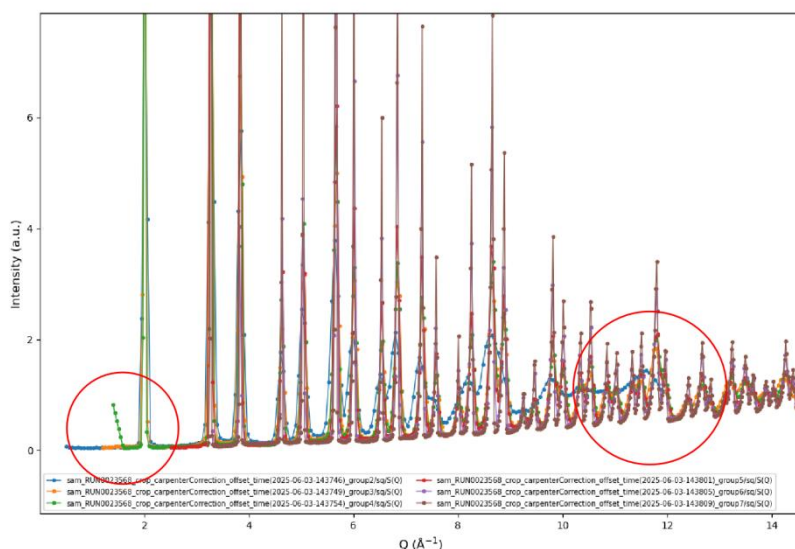


图 28 使用默认参数转的 S(Q)数据的绘图结果。一个正常的 S(Q)数据，应该保证其高 Q 处逼近 1，可以看到蓝色曲线（group2）明显有问题。还有，绿色曲线(group4)在低 Q 处存在不自然的翘起，也需要处理。

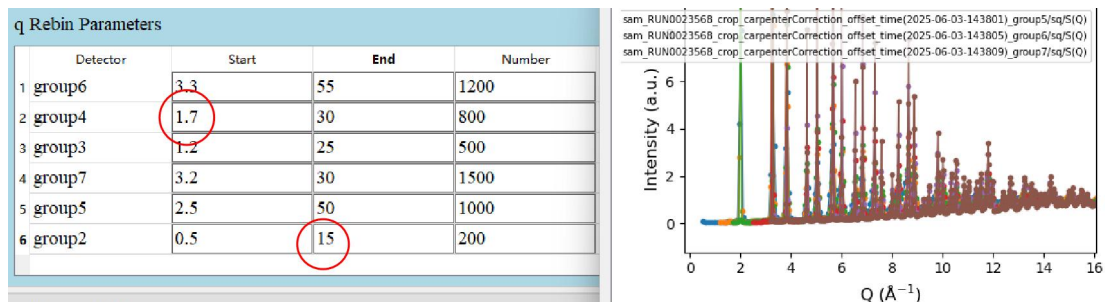


图 29 将 group4 的 q rebin start 参数调高，截掉了翘起的部分。将 group2 的 q rebin end 调高，将原本截掉的有效数据也包含进来，曲线的收敛性改善。

Step3: 将 S(Q)数据 merge 为 1 条数据

在上一步可以看到，来自不同探测器的数据其 Q 值的起止范围明显不同，对于 PDF 来说，低 Q 区的数据十分重要，但是 Q 值下限最低的 group2 的数据在高 Q 区的覆盖范围太小。因此，有必要将不同覆盖区域的数据 merge 成一条数据，然后用于 PDF 获取。

首先，展开“Merge S(Q)”模块，然后点击“Load Base Data”，选则 merge 的基底数据。然后，点击“Add”按钮，选择希望和基底数据进行 merge 的其他数据，此时，列表栏中会显示出一行数据，并且有“overlap start”和“overlap end”两个参数需要填。这两个参数决定了在 merge 这两条数据时，merge 的节点在哪个位置，在这俩参数定义的数据段之前，是基底数据，在这之后，是进行 merge 的其他数据，这段数据则经过处理使得两端数据可以平滑地接到一起。可以看出，merge 后的数据是两段数据的拼接。这之后，可以继续点击“Add”，添加其他数据，该模块会按照从上往下的顺序依次对数据进行 merge。最后，根据该模块中设置的 Q rebin 参数，产生一条新的 merge 后曲线。整个流程如图 30 所示。

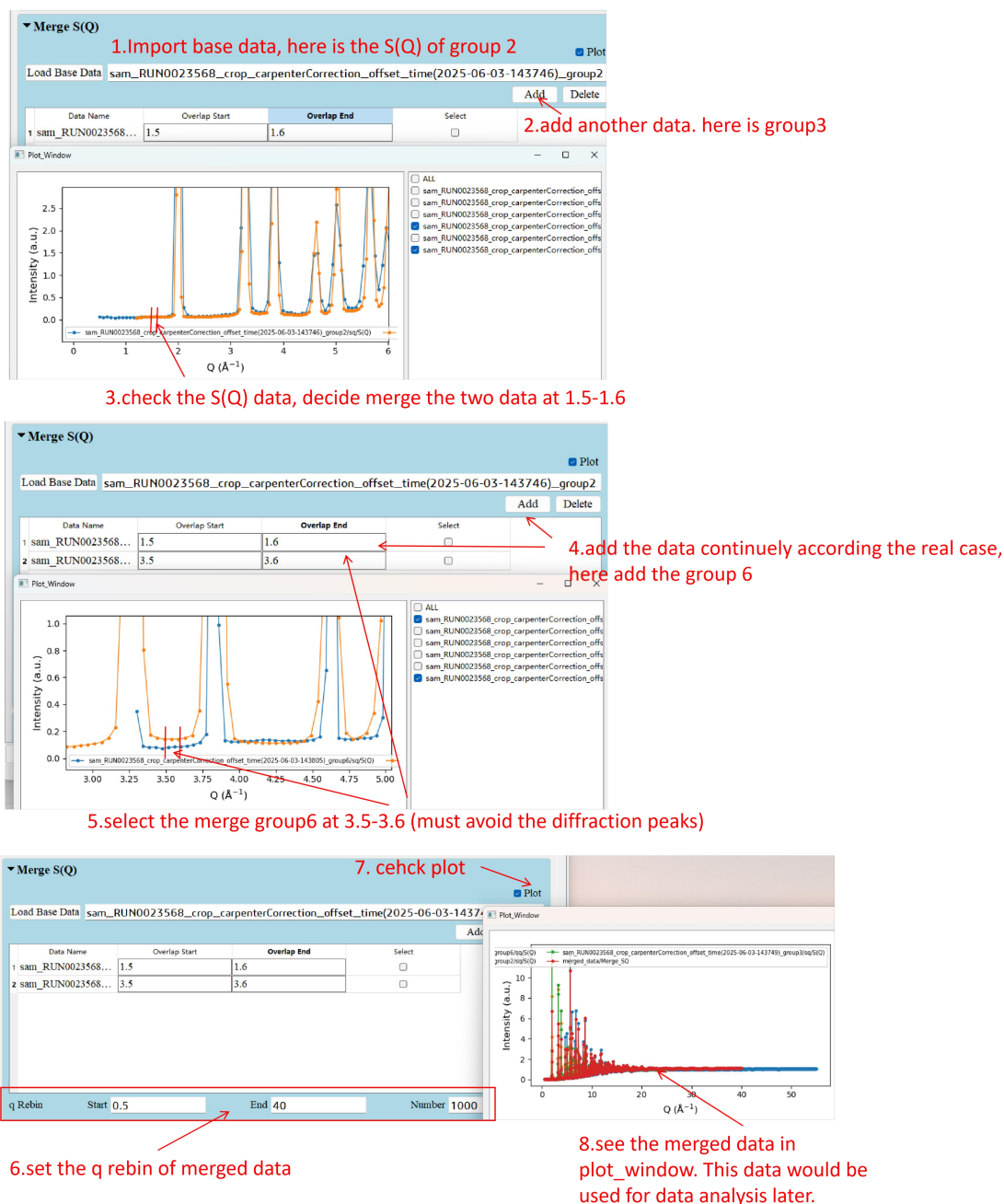


图 30 使用 Merge S(Q)模块的流程示意图

Step4. Save S(Q) (可跳过)

在 Merge S(Q)模块之后, 紧接着一个“Save S(Q)”模块。这个模块用于保存 S(Q)数据, 如果需要就展开, 然后在上面指定一个保存路径即可, 不用的话直接折叠就可以。

Step5. 配置“Get PDF”模块, 获得 PDF 数据。

如图 31(a)所示, 展开“Get PDF”模块, 然后如何 r Rebin 参数, 这三个参数决定了 PDF 数据的起止点以及数据点数量。一般需要勾选“Lorch”, 由于使用有限的 S(Q)数据转 PDF 时会造成截断效应, 导致 PDF 数据中出现周期性震荡信号, 该选项用于消除这些震荡。Self Term 用于计算 $Q_i(Q) = Q \cdot (S(Q) - \text{Self Term})$, 默

认值为 1，一般不需要改动。PDF Type 有三种，“G(r)”、“g(r)”和“RDF”。注意 PDF 数据的命名规则比较混乱，这里的 G(r)在一些地方也被称作 D(r)，是比较常用的数据（注意，在存在 D(r)的命名体系下，也存在 G(r)，但是此 G(r)和本软件中的 G(r)不同，该软件中的 G(r)在该命名体系下和 D(r)等价，请注意区分）。勾选“Plot”后，点击“Run & Plot”，绘图窗口中会出现 PDF 数据，如图 31(b)所示。

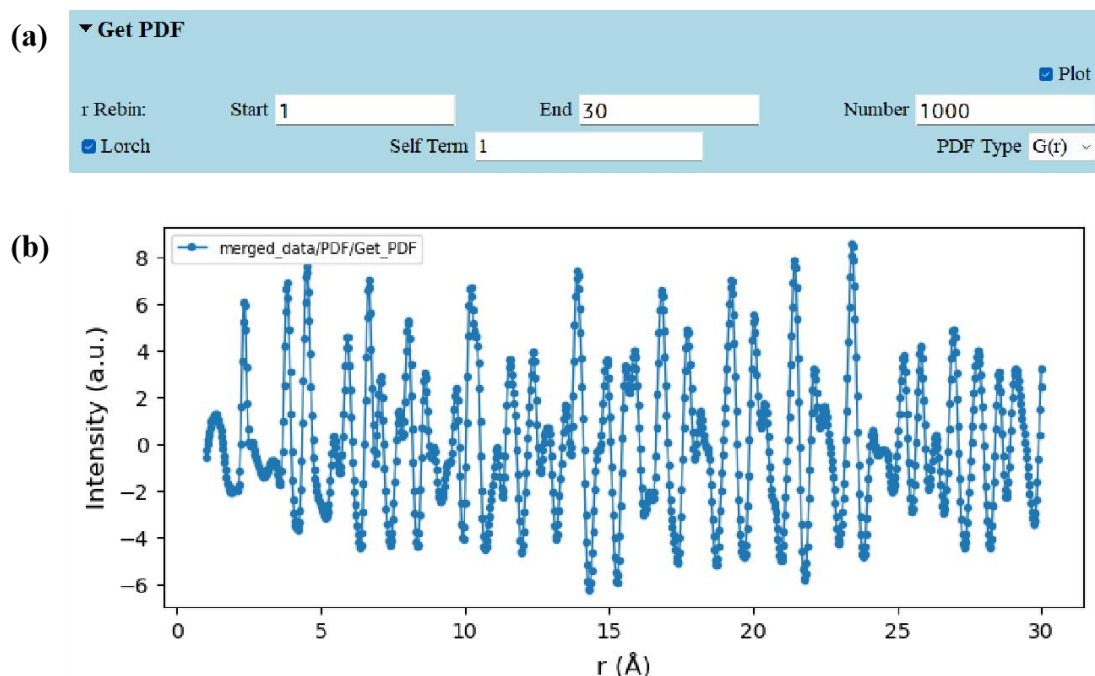


图 31 (a)设置好后的“Get PDF”界面。注意其中的 r Rebin 参数可以根据自己的实际情况调整。“Self Term”项则一般不需要改动。(b)绘制出的 PDF 数据“G(r)”。

Step6. 配置“Save PDF”模块，保存 PDF 数据。

展开最后的“Save PDF”模块，指定一个本地的保存路径即可。只要该模块配置好并且保持展开状态，点击“Run & Plot”就可以将数据保存到本地，保存的数据如图 32 所示。

```
G_r_merged_data.txt
G_r_sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143746)_group2....
G_r_sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143749)_group3....
G_r_sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143754)_group4....
G_r_sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143801)_group5....
G_r_sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143805)_group6....
G_r_sam_RUN0023568_crop_carpenterCorrection_offset_time(2025-06-03-143809)_group7....
```

图 32 保存的 PDF 数据，包括 merge 后数据的 PDF 以及 merge 前各探测器的 PDF。总之，只要在处理过程中产生了 S(Q)数据，都会被转为 PDF，其中用户需要的可能只是 merge 后数据的 PDF，需要根据需要筛选使用。

1.4.2 通过 Hermite 多项式拟合获取 PDF 数据

使用 Merge 的方法获取 PDF 数据存在一个明显的缺陷，就是不同探测器的分辨率不同，同一衍射峰在不同探测器上的计数也不同。直接将数据 Merge 在

一起，相当于忽略了这些因素造成的影响。RZera 提供了另一种获取 PDF 的方法，通过“Hermite 多项式拟合”的方式得到无仪器展宽影响的 PDF 数据。该方法由 Martin T Dove 发展，在他的帮助下整合入了 RZera，提供给散裂中子源的用户使用。其背后的理论不在此介绍，仅说明使用方法。

Step1: 切换到“PDF Hermite”页面并完成数据规约以及转 $S(Q)$

用户首先需要切换到“PDF Hermite”页面，然后会发现该页面最开始的几个模块和“Reduction”页面中仍然是完全一样的。这是因为在获取 PDF 数据之前要先对数据进行 reduction。这里，知道“Division”模块，其操作方式和第一节介绍的是完全一样的，这里不做赘述。这之后，后面的“Convert to $S(Q)$ ”模块和该节第一部分“通过 Merge 获得 PDF 数据”中的使用方法完全一样，这里也不做赘述。

Step2: 配置 Hermite Fitting 模块

展开 Hermite Fitting 模块后，填写其中的内容，如图 33 所示。首先需要填写 q Rebin，注意，这里的 q Rebin 应该涵盖所有 $S(Q)$ 数据中希望参与拟合的部分。比如，这里我所有探测器的 $S(Q)$ 数据在 0.5-55 的范围内，我希望用所有这些数据进行拟合，那么我的 q Rebin 中的 Min 和 Max 应该写 0.5 和 55，如果我写了 0.5 和 50，那么 group7 中大于 50 的部分数据将不会参与拟合。r Rebin 根据自己的需求填写即可。在该模块“Get and Preprocess $qI(q)$ ”子栏目中，Order of Hermite 一般默认选择“Auto”即可。Self Term 默认为 1 即可。“Chebyshev Fitting”正常需要勾选。“Fitting Order”默认为 8，一般不需要修改，如果修改，必须改为一个偶数。如果样品是晶体，需要勾选“Crystal”，如果是非晶则不需要勾选。“Lorch”正常需要勾选。这里的“Plot”并非用于画 PDF 数据，而是用于画预处理后的 $qI(q)$ 数据。用户可以在进行 PDF 拟合前，先检查 $qI(q)$ 数据是否正常，这时用户可以不勾选下面的“Hermite Fitting”，然后点击“Run & Plot”，查看 $qI(q)$ ，如图 34 所示。

确认 $QI(Q)$ 数据正常后，勾选“Hermite Fitting”。如果也勾选了“Fitting with Instrument Resolution”，在拟合时会卷积进入仪器参数，使得最后得到的 PDF 是从仪器参数解耦合的数据。

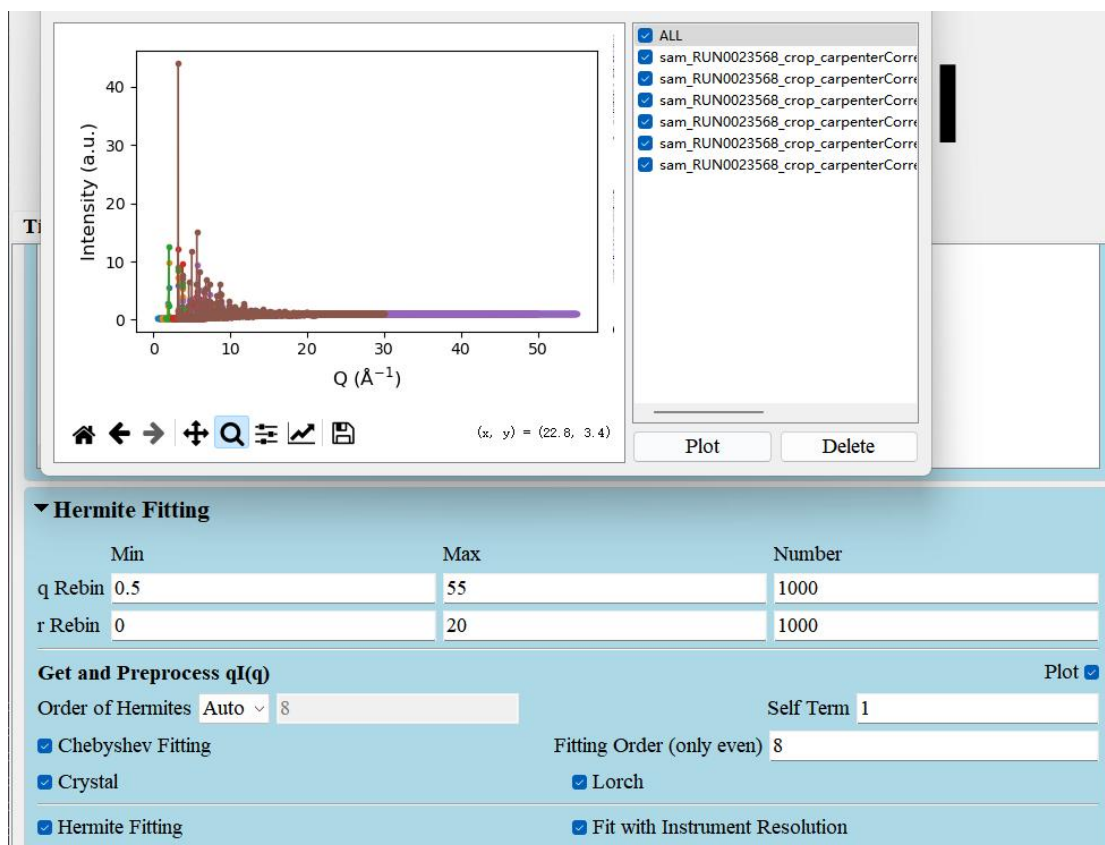


图 33 配置好的 Hermite Fitting 模块。

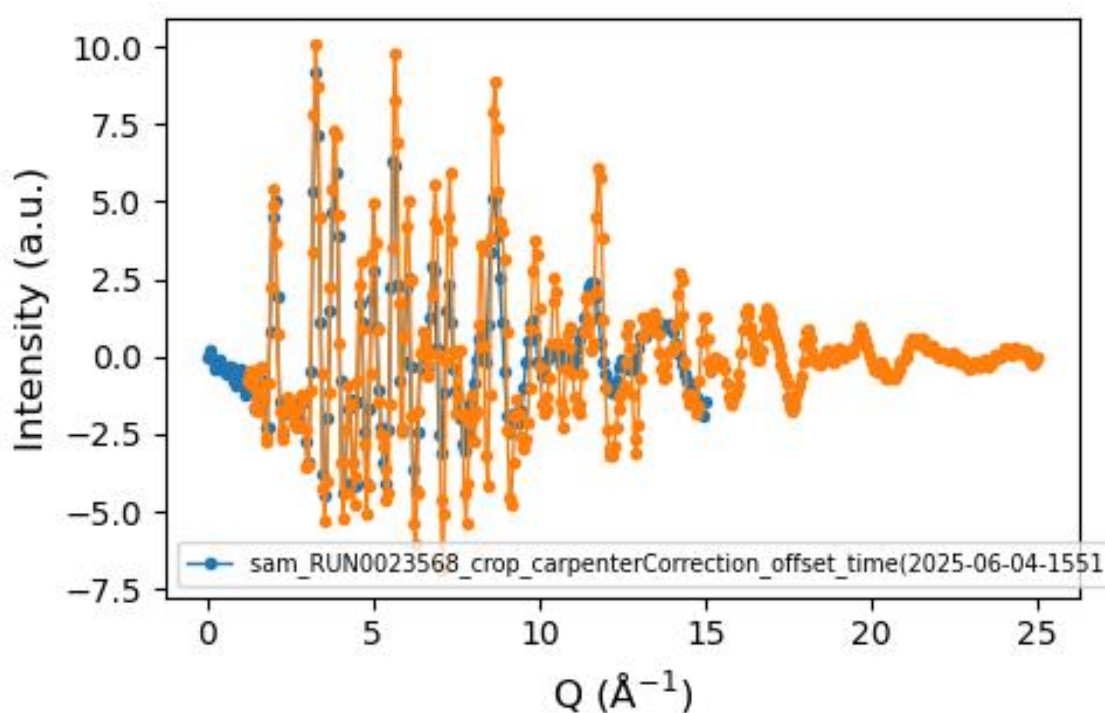


图 34 处理后的 $qI(q)$ 数据。评估此数据的要点是其在低 Q 处是否趋向 0，以及其在高 Q 处是否也趋向 0。

Step3:配置最后的 Save PDF 模块。

在该页面的最后，有一个“Save PDF”模块，如图 35(a)所示。此模块中设置了 Plot 和 PDF 类型，如果勾选了 Plot，点击“Run & Plot”，会把 Hermite 拟合得到的 PDF 结果绘制出来，如图 35(c)所示，绘制的 PDF 类型则和该模块中指定的相同。同时，如果制定了 Save Path，该模块会把 PDF 数据保存到指定路径下，如图 35(b)所示。

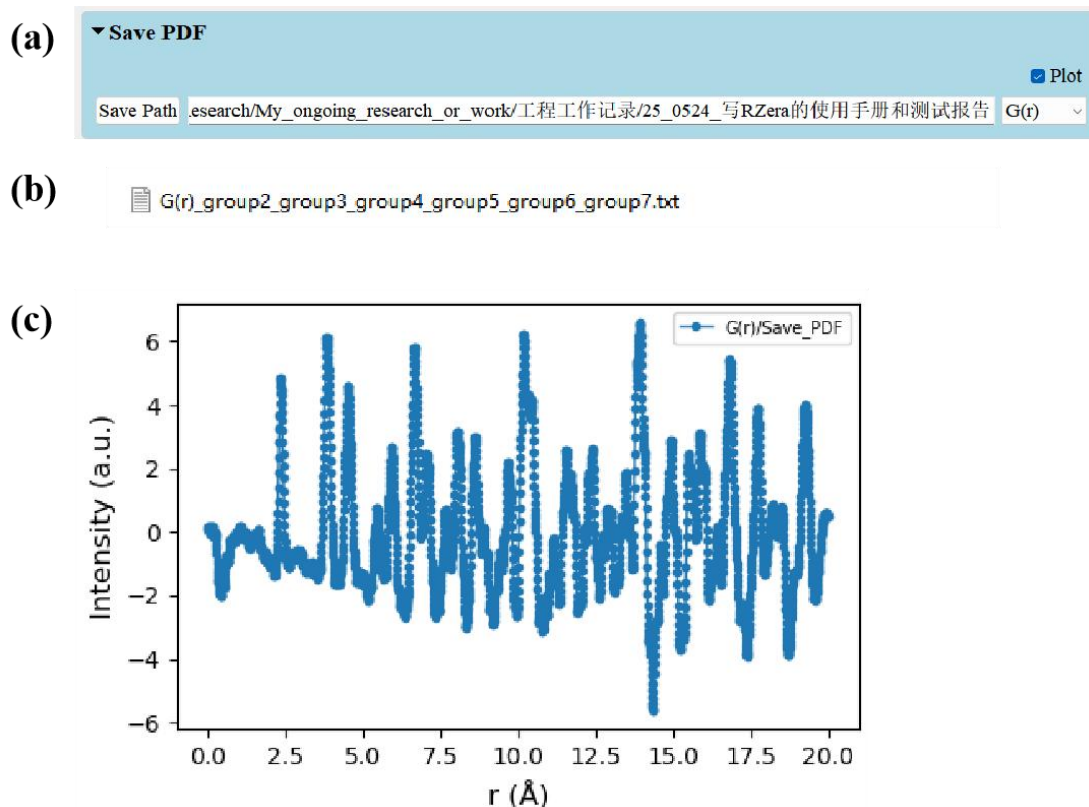


图 35 (a)配置好的“Save PDF”模块。(b)保存的 PDF 数据。(c)绘出的 PDF 数据 $G(r)$

Step4(可跳过):配置“Get S(Q) from PDF”模块和“Save S(Q)”模块

这两个模块的目的是从最后拟合的 PDF 数据逆向计算出其对应的 S(Q)数据。这俩模块存在的原因是使用 Hermite 拟合获取 PDF 时，直接从各探测器的 S(Q)出发，不需要 merge，因此不会产生其对应的 S(Q)数据。考虑到后续数据分析可能使用到，因此增加了这两个模块。

如图 36(a)所示，配置好 q Rebin 和 Save Path 即可。这里 q Rebin 建议和“Hermite Fitting”模块中的 q Rebin 保持一致。计算得到的 S(Q)数据如图 36(b)所示。保存的数据如图 36(c)所示。

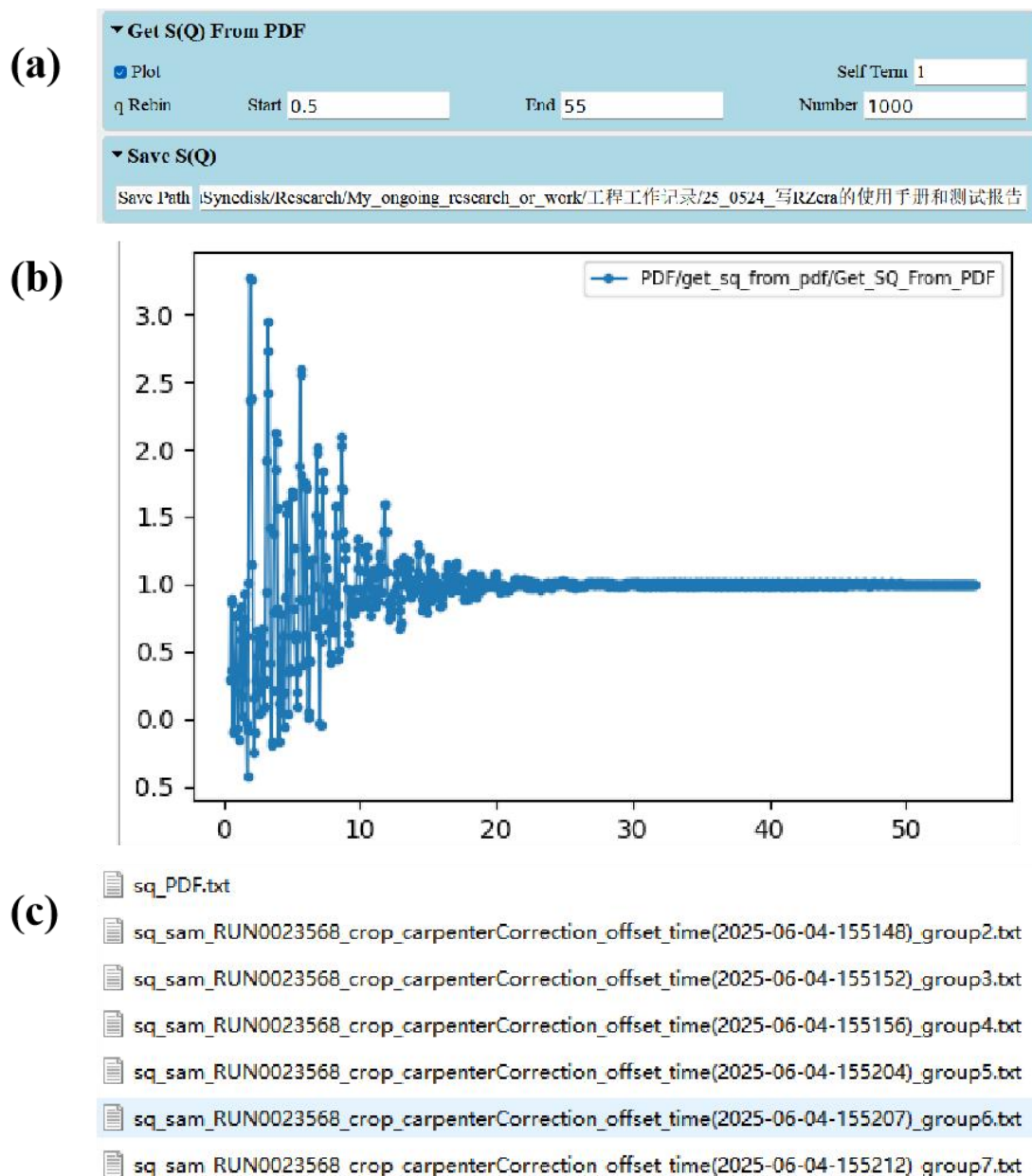


图 36 (a)配置好的“Get S(Q) from PDF”模块和“Save S(Q)”模块。(b)计算得到的 S(Q)数据。可以看到，其和真实 S(Q)数据的一个明显差距是，其存在较明显的周期震荡，这点必须注意。(c)保存的 PDF 数据。其中“sq_PDF.txt”是计算的 S(Q)数据，其他的则是不同探测器的真实 S(Q)数据，这些数据会一并保存。

1.5 获取 offset 文件的简易操作指南(当前仅支持 BL16)

Offset 文件是在对数据进行 Time Focusing 时一般会使用到的文件，一般谱仪会提供此文件。这里介绍如果没有 offset 文件，应该如何去获取。offset 文件的获取涉及到很多参数的调节，是一个比较复杂的过程，但是 RZera 提供了包含已经调节好参数的配置文件，用户只需要导入配置文件即可。下面介绍如何操作：

1.5.1:打开 offset 界面

在谱仪归约界面的左上角，有一个 other 选项，点击后在下拉框中有“offset_correction”选项，点击进去后就可看到如图 37 所示的界面。

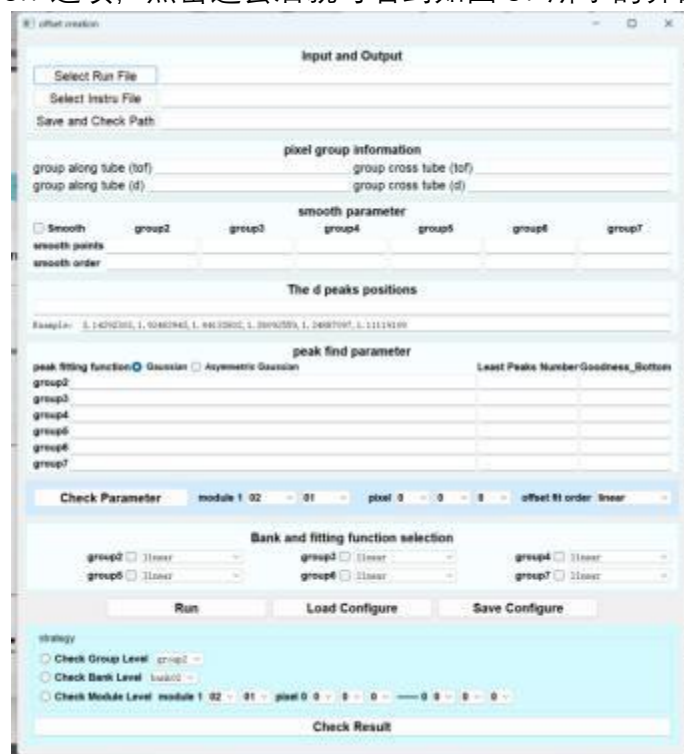


图 37 获取 offset 文件的操作界面

1.5.2:导入配置文件

如图 38 所示，点击“Load Configure”，选择 RZera 提供的配置文件导入即可。

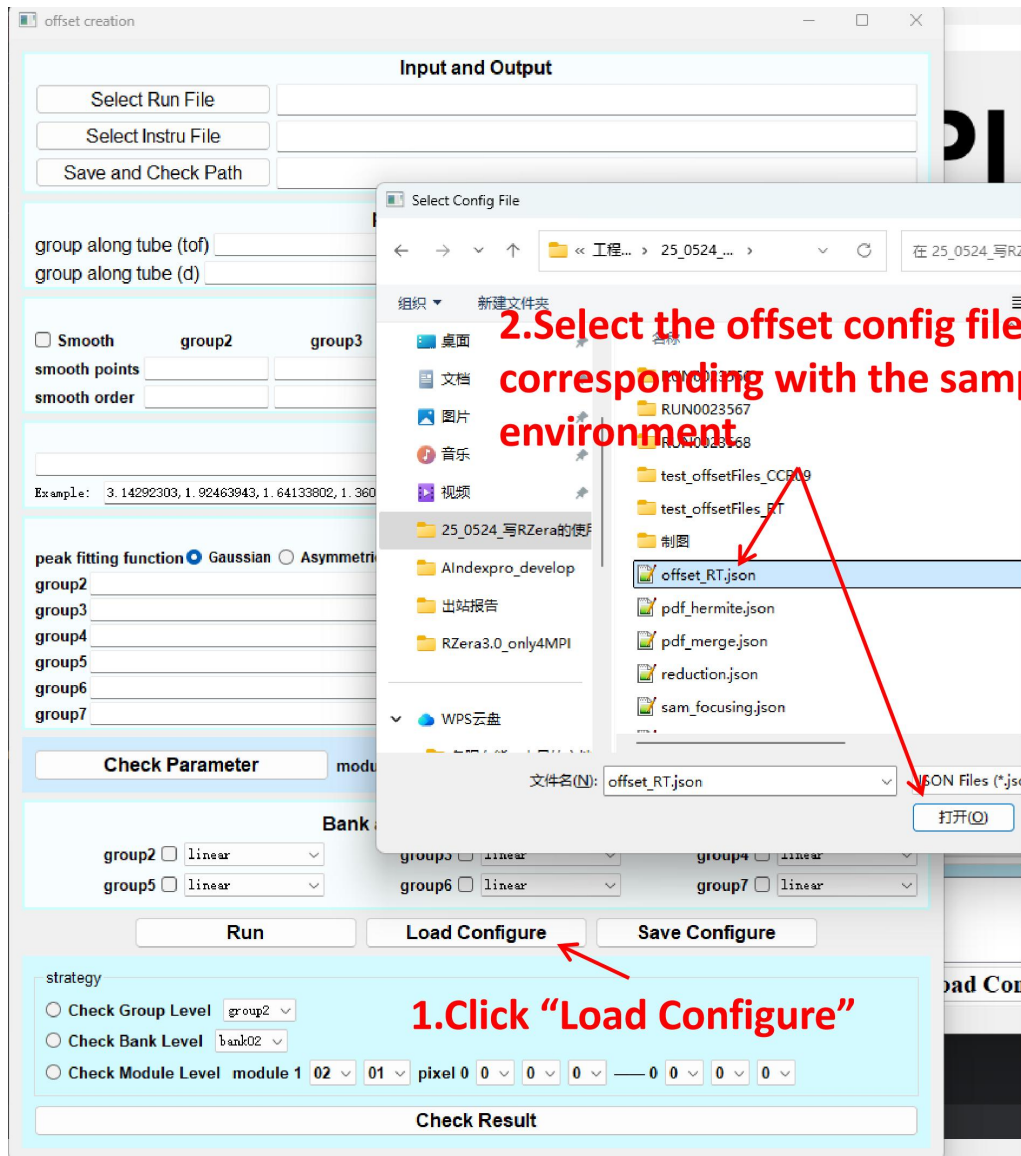


图 38 导入配置文件的方法

1.5.3:修改 3 个路径

界面的最上方有 3 个路径。第一个是进行 offset 修正使用的 Si 标准样品所在的路径(注意，必须是 Si，不能是任何其他样品，否则需要修改其他参数)；第二个是 pidinfo 文件夹所在的路径（该文件夹已经随软件提供）；第三个是保存产生的 offset 文件的文件夹路径。使用者需要确保这三个路径是正确的，如图 39 所示。修改好路径后，使用者可以点击“save configure”将包含正确路径的 json 文件保存下来，后续使用就不需要再修改路径了。

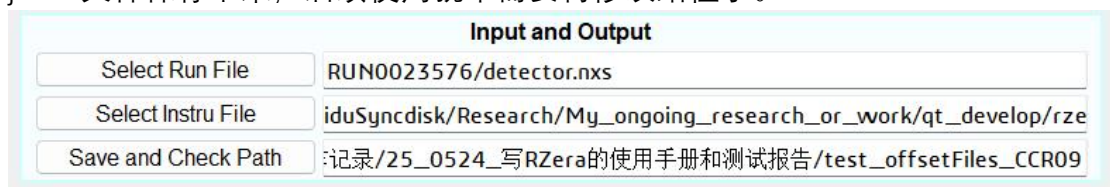


图 39 3 个需要检查的路径。点击“Select Run File”，导入作为 offset 基准的 Si 实验数据。点

击“Select Instru File”，导入仪器像素位置文件。点击“Save and Check Path”，指定 offset 文件的存储路径。

1.5.4:点击 Run 按钮，等待文件获取完成

点击“ RUN”后，会弹出进度条，如图 40 所示。进度条读条完成后，offset 文件就生成完毕了，使用者可以从上一步中自己指定的保存 offset 文件的路径中得到这些文件。

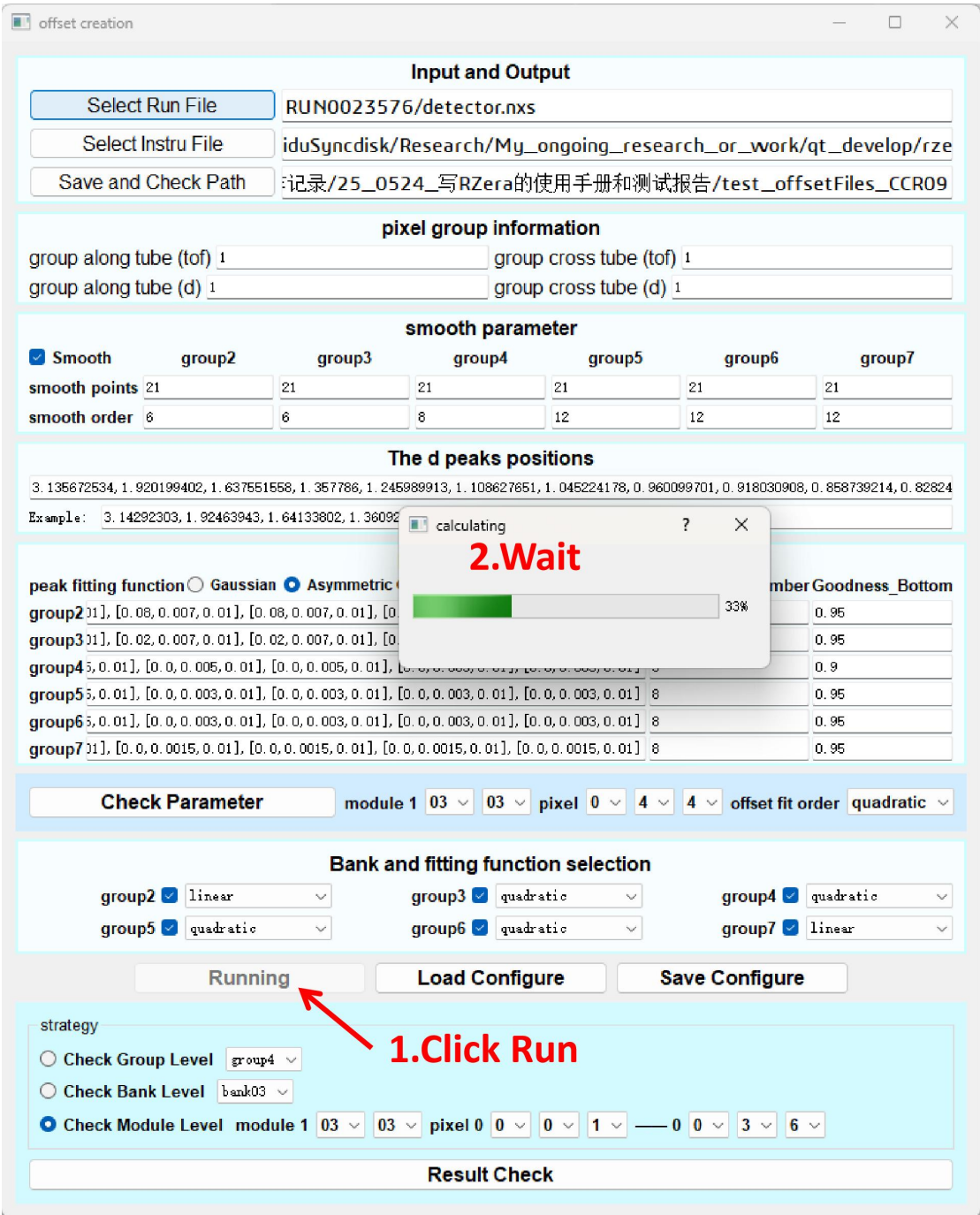
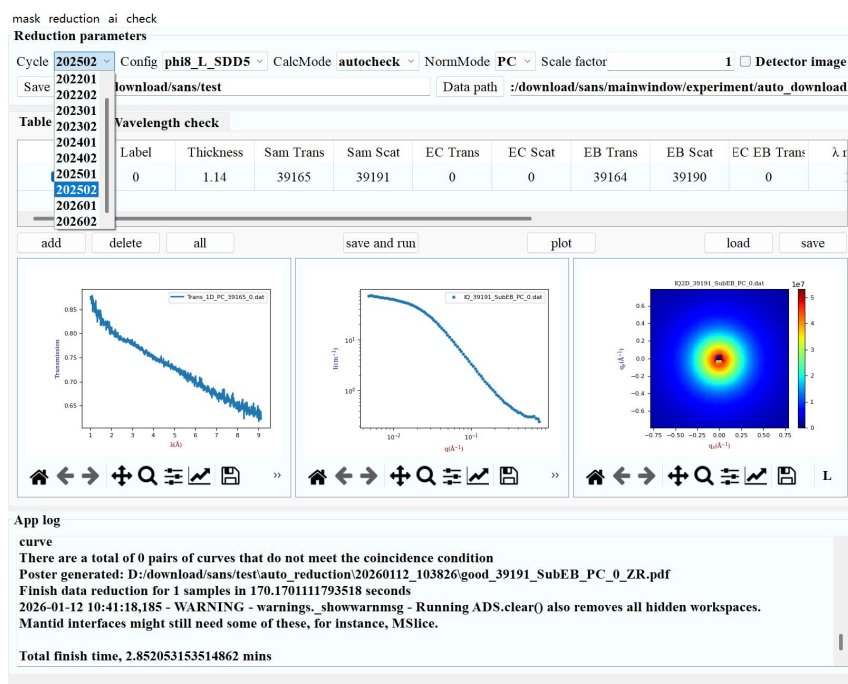


图 40 点击 Run 后等待 offset 文件的生成

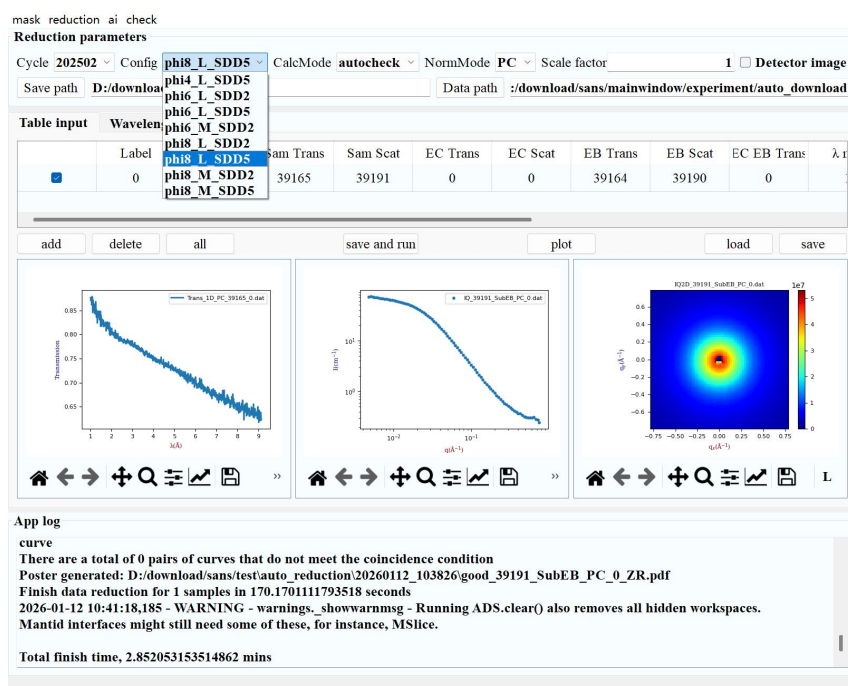
2. 小角散射谱仪软件操作指南

2.1 规约界面功能介绍

1、选择 **Cycle**: 谱仪仪器参数按年份选择，一年分为上下两部分，在选项中体现为 01 和 02 两部分。例：202501，代表谱仪仪器参数是 2025 年上半年刻度的。



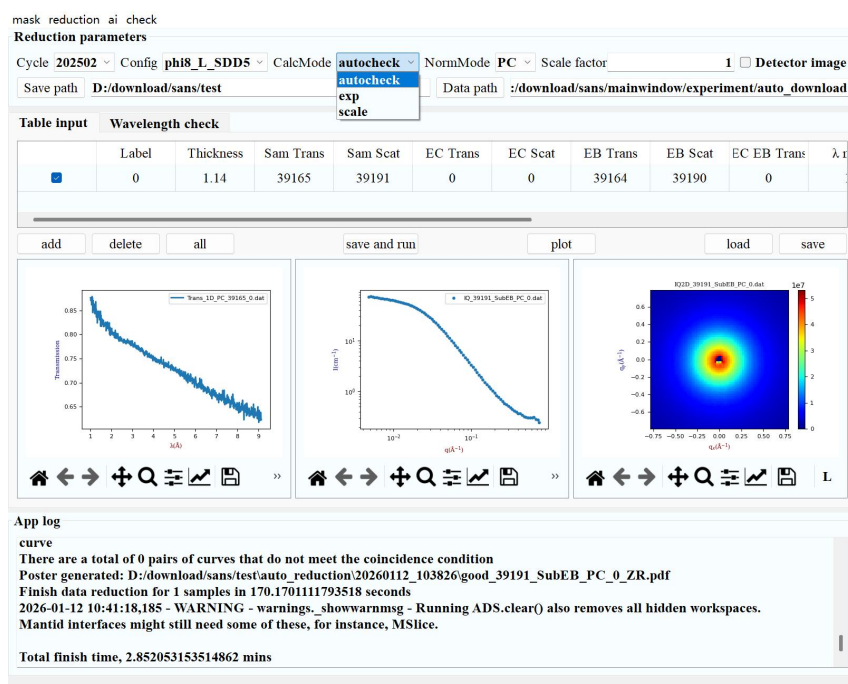
2、选择 **Config**: 该参数决定了软件使用的仪器刻度文件夹。请选择正确的探测器信息。



3、选择 **CalcMode**: 计算模式。

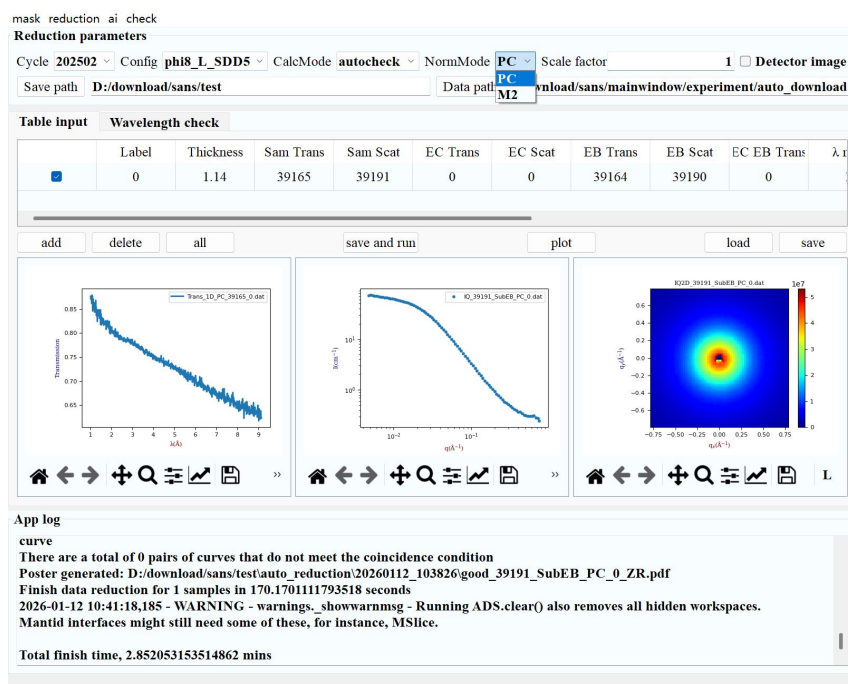
- scale: scale 因子计算模式，在这个模式下，只能勾选一个实验数据。

- exp: 规约模式，完成所有勾选实验的规约。
- **autocheck**: 自动完成 bragg 边判断，取向判断，使用合适的波长范围和积分角度做自动规约。将判断结果保存为 pdf 文件，用于人工校检，大幅提高规约效率。**推荐使用该模式**



4、选择 NormMode: 归一化选择。

- PC: 使用中子流强归一。
- M2: 使用监视器归一。

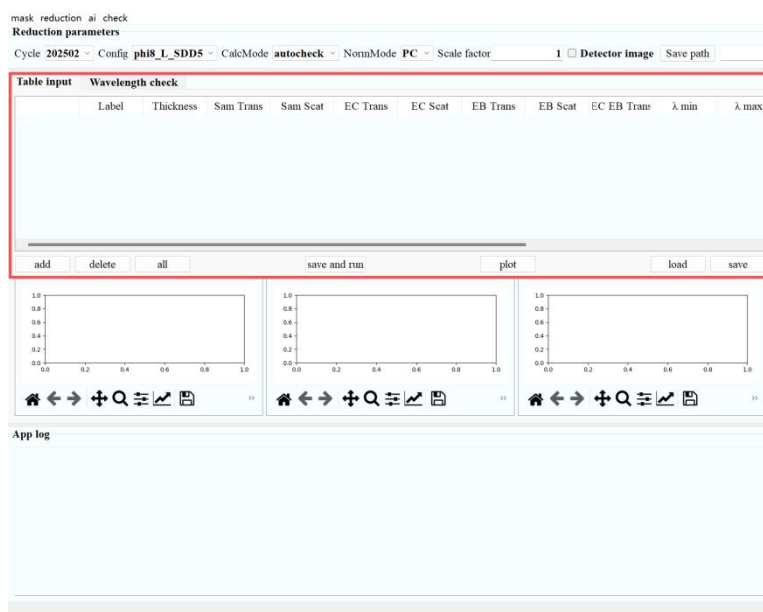


- 5、Scale factor: ref 数据/exp 数据，由 scale 模式计算得到。
- 6、Detector image: 勾选以保存计算过程中的探测器二维图像。不勾就不存。
- 7、Save path: 计算结果的保存路径。

8、**Data path**: 数据文件的路径。

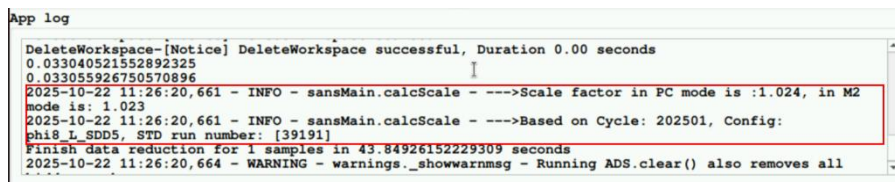
9、实验数据填写

- **load**: 导入包含样品厚度、实验、盒子、背底 RUN 号，波长范围的 excel 表格。
- **save**: 保存表格为 excel 格式。
- **add**: 在表格最后一行后面添加空行。
- **delete**: 删除勾选行。
- **all**: 勾选或取消勾选全部行。
- **save and run**: 进行一次计算，将产生的文件保存至 save path 中。1D、2D、Trans 文件夹保存了绘图和分析的结果。可根据 Sam Scat 和 Sam Trans 的 RUN 号找到对应文件。
- **plot**: 绘制所有勾选实验的透过率曲线、一维散射曲线（双 log）、二维散射图。注意：二维图片只绘制一张，如果多个勾选，只显示最后一个勾选的图片。



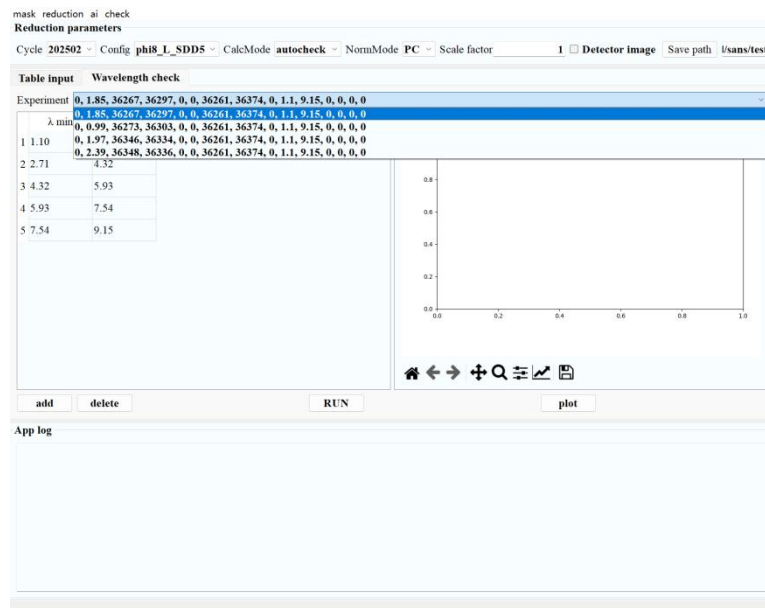
10、**App log**: 记录了计算过程中的细节。请在完成一次计算后，确认是否输出错误信息，任何错误信息的存在，可能导致计算结果不正确。

注意：在 scale 模式下，计算得到的 scale 因子也被记录在此（图中红框），请根据输出将结果填入 **Scale factor** 中。



2.2 wave slice 界面功能介绍

1、**Experiment**: 获取 Table input 页面，表格中的实验信息。选择需要 wavelength check 的实验。



2、waveslice 参数

- **add**: 在波长区间内添加一个分段。根据第一段的开始波长、最后一段的结束波长，等长的设置分段。
- **delete**: 在波长区间内减少一个分段。根据第一段的开始波长、最后一段的结束波长，等长的设置分段。
- **RUN**: 开始计算，并将 waveslice 的结果保存至 Save path 下的 check 文件夹中。
- **plot**: 根据波长切分的情况绘制分段图片。

2.3 规约参数控制面板

从 **reduction** 菜单进入：弹出专家参数面板，这里可以对规约过程中的细节参数做出调整。

Input File

Mask path Mask name

Ref. path Ref. name

Input Params

Scale factor Qstep

Qbin Wave rebin 2D

Qxy max Qxy step

Trans mode Smooth pt

Output Data

☒ data 1D ☒ data 2D

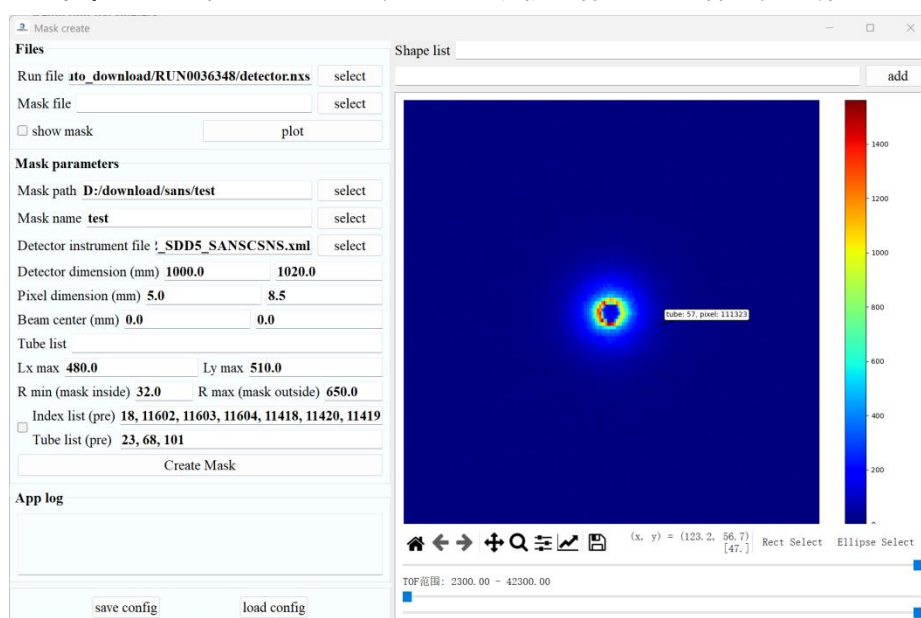
- 1、**Mask path**: 点击 select，选择希望使用的 mask 文件。
- 2、**Ref path**: 点击 select，选择希望使用的 Ref 文件，该文件被用于 Scale factor 的计算。

3、Trans mode:

- smooth: 透过率曲线做平滑修正
 - raw: 使用未修正的透过率曲线
- 4、smooth pt: 平滑后透过率的点数。
- 5、Save configure: 保存修改结果。
- 6、Qstep: 单位转换为 q 时，一维散射曲线的步长。
- 7、Qbin: 单位转换至 q 时，组成一维散射曲线的点数。
- 8、wave rebin 2D: 二维探测器图像波长范围。
- 9、Qxy max: 单位转换至 q 时，二维小角散射图像的最大值。
- 10、Qxy step: 单位转换至 q 时，二维小角散射图像的步长。

2.4 创建 mask 文件

从 mask 菜单进入: 弹出 mask 面板，这里可以根据特定 RUN 样品，制作 mask 文件。



1、Files 区域:

- Run file: 选择需要绘制的 RUN 文件。
- Mask file: 选择已有的 Mask 文件 (第一次做没有 mask 文件可以空着)
- show mask: 是否绘制 mask 情况
- Plot: 画出二维图片。

2、mask parameters 区域:

- Mask path: 选择需要保存 mask 文件的路径。
- Mask name: 输入希望保存的 mask 文件的名称。
- Detector ins file: 探测器的参数文件。
- Detector dimension: 探测器的实际大小。
- Pixel dimension: 像素大小。
- Beam center: 光斑中心。
- Tube list: 需要 mask 的管子号。
- Lx max 和 Ly max: 红色区域被 mask。示意图如下
- R min 和 R max: 以光斑中心为圆心，mask 掉 R min 以内和 R max 以外的区域。
- Index list (pre): 预先设好的需要 mask 的像素。

- tube list (pre) : 预先设好的需要 mask 的管子 (如需修改, 请勾选)
 - Create Mask: 开始创建并保存 mask 文件。
- 3、**绘图区域:**
- Rect Select: 划动鼠标框选需要 mask 的矩形区域。
 - Ellipse Select: 划动鼠标框选需要 mask 的椭圆形区域。
 - 右上区域, 实时更新当前使用 Rect Select 和 Ellipse Select 选择的区域, 点击 add, 将区域填充至 Shape list 中。
 - Shape list: 需要 mask 的自定义矩形和椭圆形区域汇总。
 - 移动滑块 1: 改变最大 counts 值, 以凸显坏点的视觉表现, 协助创建 mask。
 - 移动滑块 2, 3: 改变 tof 加和范围, 帮助发掘异散射异常信号。

2.5 自动规约检查功能控制面板

从 **check** 菜单进入: 弹出 check param 面板, 这里可以调整检查 brag 边与取向的判据。

The screenshot shows a software interface for adjusting parameters. It is divided into three main sections, each with a title and several input fields:

- Transmittance**
 - Brag threshold: 0.04
 - Min threshold: 0.7
- Orientation**
 - Residual standard: 0.4
 - Count threshold: 5
 - Angle range: 30.0
 - Q grid: 82
 - Radius: 10
- Wave slice**
 - Deviation threshold: 0.2
 - Accept ratio: 0.8

- 1、**Transmittance:** 从峰值和透过率下限判断是否存在 brag 边。
 - Brag threshold: 对透过率做二次曲线拟合, 拟合优度大于 Brag threshold, 被认为其不具有布拉格边。Brag threshold 被设置得越大, 越不容易忽略布拉格边。
 - Min threshold: 设置透过率被允许的最小值。
- 2、**Orientation:** 以光斑为中心的圆弧做强度-角度曲线, 通过峰值判断是否存在取向。
 - Residual standard: 对以光斑为中心的圆弧做强度-角度曲线做一次拟合, 计算残差, 残差通过一次拟合曲线归一, 存在超过 Residual standard 的峰值, 则认为存在取向。该值越大, 越不容易被判断为有取向。
 - Count threshold: 排除少量离群的点对判断结果的影响, 超过 Residual standard 的离群点必须大于 Count threshold 才会被判断为存在取向。
 - Angle range: 通过经验判断的沿角度的积分宽度, 这个代表对沿取向方向多大的扇形角度进行积分, 计算得到一维 Iq 曲线。
 - Q grid: 对二维散射图样做插值, 插值的点数。
 - Radius: 以光斑为中心, 所做的判断圆弧的半径, 单位为无量纲, 代表点数。
- 3、**Wave slice**
 - Deviation threshold: 离群点判据, 代表偏离当前对比曲线平均值的百分比。
 - Accept ratio: 合群点的数量占总点数的百分比大于 Accept ratio, 被认为是重合度较好的结果。

2.6 规约结果介绍

Output directory		
	Configuration_initial.json	} Input Files
	Sample_info.xlsx	
	1D IQ_runno_SubEC/EB_PC/M2_batch.dat	Sample Iq after correction and bkg subtraction
	1DRaw IQ_runno_Raw_PC/M2_batch.dat	Sample and bkg scattering curve after correction
	2D IQ2D_runno_SubEC/EB_PC/M2_batch.dat	Sample 2D data after correction and bkg subtraction
	2DRaw IQ2D_runno_Raw_PC/M2_batch.dat	Sample and bkg 2D scattering curve after correction
	Trans Trans_all/1D_PC/M2_runno_batch.dat	Transmission curve
	Check IQ_runno_check_no.dat	Iq with multi-wave check
	DetectorImage RUN00runno_detector_image.png	2D detector picture before correction
	logs sans_reduction_date.log	log
		runno_trans_check.png
	auto_reduction Quality_runno_SubEC/EB_PC/M2_batch.dat	the result of auto check for reduction